English

(1001CJA106216230186)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
08-05-2024

JEE(Main+Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

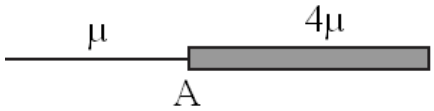
SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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1. A taut string is made of two segments. To the left of point 'A', it has linear mass density of μ kg/m and to the right of A, its linear mass density is 4μ kg/m. A sinusoidal wave pulse is travelling towards right.



Which of the following is correct ?

(A) Ratio of amplitude of reflected and transmitted wave is $\frac{1}{4}$

(B) Ratio of amplitude of reflected and transmitted wave is 2

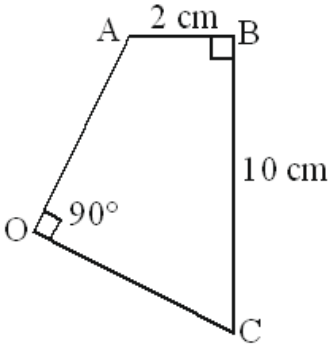
(C) Ratio of transmitted power and reflected power is 8

(D) Ratio of transmitted power and reflected power is 16

2. One mole of a monoatomic gas is heated in such a way that its molar specific heat is 2R. During the heating, the volume of the gas is doubled. By what factor does the temperature of the gas change ?

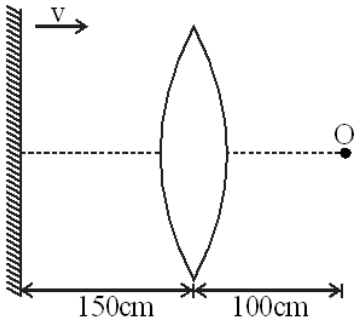
(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) 4

3. The figure shows a uniform slab of mass 7 kg. Find the moment of inertia about the axis normal to the plane of drawing and passing through the point O. Given OA = OC, BC = 10 cm and AB = 2cm.



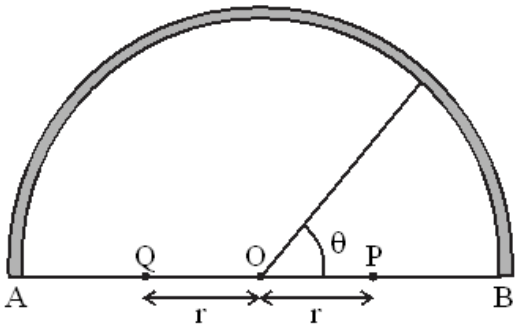
(A) 84 kg cm^2 (B) 168 kg cm^2 (C) 188 kg cm^2 (D) 198 kg cm^2

4. An observer is standing at a point O, at a distance of 100 cm from a convex lens of focal length 50 cm. A plane mirror is placed behind the lens at a distance of 150 cm from the lens. The mirror now starts moving towards right with a velocity of 10 cm/s. What will be the magnitude of velocity of her own image as seen by the observer, at the moment when the mirror just starts moving ?



(A) $\frac{200}{9}$ cm/s (B) $\frac{200}{3}$ cm/s (C) $\frac{20}{9}$ cm/s (D) $\frac{10}{9}$ cm/s

5. Charge distribution on a rod bent as a semicircular arc of radius R follows a function $\lambda = \lambda_0 \sin\theta$. Here λ is line charge density at a point on the rod, λ_0 is a positive constant and θ is angular position of the point as shown in the figure. V_0, V_P and V_Q represents the electrostatic potential at points O (centre point), P and Q respectively as shown in figure. Which of the following is correct?



(A) $(V_P = V_Q) < V_0$ (B) $(V_P = V_Q) > V_0$

(C) $V_P = V_Q = V_0$ (D) $V_P > V_Q > V_0$

6. Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ. If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. Find N at time $t = t_0$.

(A) $\frac{2t_0 - 2\lambda t_0^2}{\lambda^2}$

(B) $\frac{2t_0 - \lambda t_0^2}{\lambda^2}$

(C) $\frac{t_0 - \lambda t_0^2}{\lambda^2}$

(D) $\frac{2t_0 - \lambda t_0^2}{2\lambda^2}$

SECTION-I(ii) : (Maximum Marks: 24)

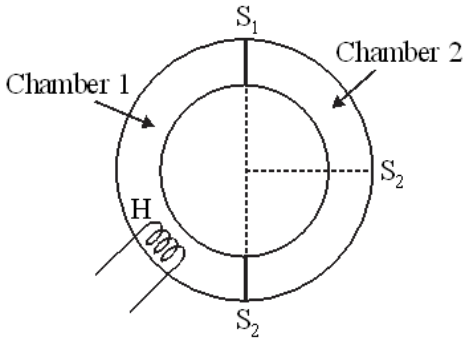
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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7. A ring shaped tube has uniform cross sectional area and its entire volume is $2V_0$. The tube is well insulated from the surrounding. Inside the tube there is an adiabatic fixed wall S_1 and another movable adiabatic partition S_2. Initially, the movable partition is diametrically opposite to S_1 and the two halves of the tube have equal amount of an ideal gas ($\gamma = 1.5$) at same pressure P_0. Now, a heater H is switched on which supplies heat slowly to one of the chambers. Heater is kept on till the partition S_2 moves through the quarter of the circle. At this position the heater is switched off and the partition S_2 remains in equilibrium. Neglect any friction as well as heat loss to the surrounding through the walls of the tube. Which of the following options are correct ?



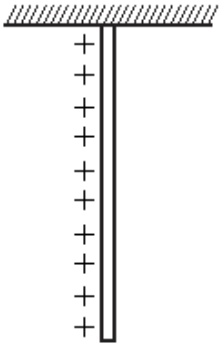
(A) Final pressure in chamber '2' is $2P_0$

(B) Final pressure in chamber '2' is $2\sqrt{2}P_0$

(C) Work done on the gas in chamber '2' by the gas in chamber '1' is $(2\sqrt{2}-1)P_0V_0$

(D) Heat supplied by the heater to the gas is $4P_0V_0(2\sqrt{2}-1)$

8. A uniformly charged non conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is λ C/m and it has a uniformly distributed mass of M. Length of the rod is L. A uniform horizontal electric field (E) is switched on in the region. Select the correct option/options :-



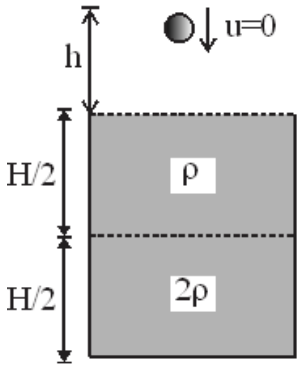
(A) If $E = \frac{mg}{\lambda L}$, then rod just manages to make it horizontal

(B) If $E = \frac{mg}{2\lambda L}$, then rod just manages to make it horizontal

(C) If $E = \frac{Mg}{\lambda L}$, the rod has maximum kinetic energy at an angle 45° with vertical

(D) If $E = \frac{Mg}{2\lambda L}$, the rod has maximum kinetic energy at an angle 45° with vertical

9. A small ball of mass m is released from rest from a height h above the liquid surface. The density of ball is $\frac{\rho}{3}$: (assume no splash)



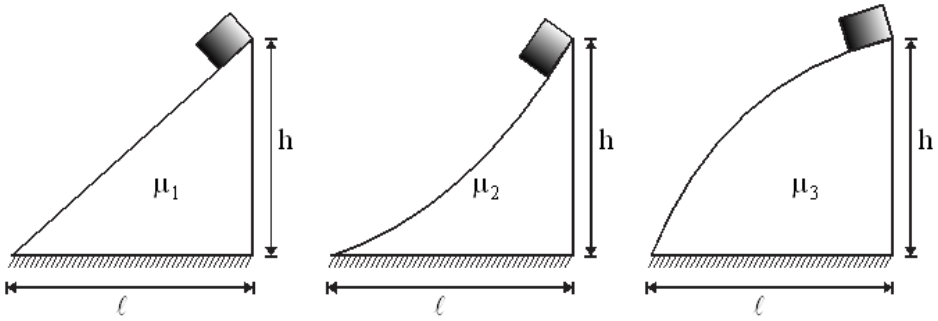
(A) If $h \geqslant H$, then the ball will be able to enter the liquid of higher density

(B) If $h \geqslant \frac{7H}{2}$, then the ball will be able to strike the bottom of the vessel

(C) If h = 2H, then the ball will cross the interface of two liquids with kinetic energy = mgH

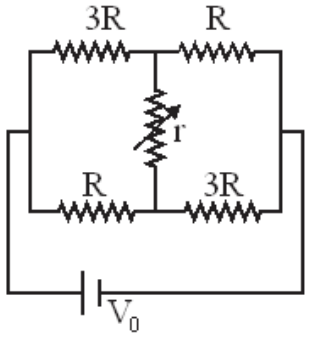
(D) If h = 0 then ball will float on the surface.

10. Consider three fixed surfaces shown in the figure. Three blocks each of mass m are released from rest from top of the three surfaces. All blocks reach ground with same speed. Length of path travelled by the blocks is same for second and third surface. If coefficient of friction of three surfaces are μ_1, μ_2 and μ_3 respectively then



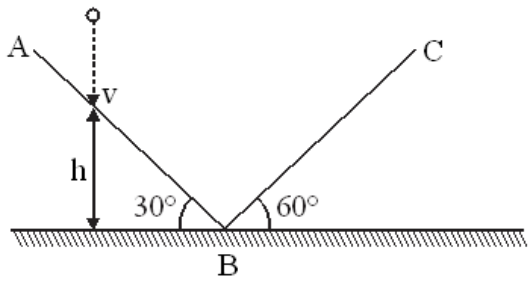
(A) $\mu_1 = \mu_2$ (B) $\mu_1 > \mu_2$ (C) $\mu_2 = \mu_3$ (D) $\mu_2 < \mu_3$

11. In the circuit shown in figure R and V_0 are constant, while r can change, taking up all values. What can be the equivalent resistance of the circuit ?



(A) 1.6 R (B) 1.4 R (C) 1.8 R (D) 2.2 R

12. There are two inclined planes, as shown in the figure. A ball strikes the inclined plane AB vertically with velocity $v = \sqrt{24\,g}$ at height $h = \left(1 + 2\sqrt{3}\right)$ m. The collision is inelastic. After this the ball moves such that it collides with the inclined plane BC perpendicular to it. Select the option(s).



(A) The co-efficient of restitution between the inclined plane AB and ball is $\frac{1}{\sqrt{3}}$

(B) The co-efficient of restitution between the inclined plane AB and the balls is $\frac{1}{3\sqrt{3}}$

(C) The time interval between two collision is $\sqrt{\frac{8}{g}}$ s

(D) The velocity of collision with the inclined plane BC is $\sqrt{6\,g}\left(1 + \frac{1}{\sqrt{3}}\right)$ m/s

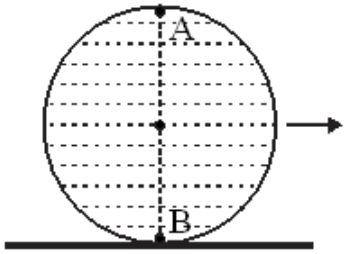
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

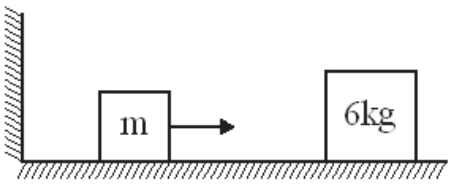
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

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1. A non-viscous incompressible liquid of mass m is filled fully inside a thin uniform spherical shell of mass m and radius R performing pure rolling on a rough horizontal surface. There are two point A and B inside the liquid on the vertical diameter separated by 2R as shown in the figure. Pressure difference between B and A is $(P_B - P_A)$. At the given instant velocity of centre of mass of given system is V_0 and kinetic energy of this system is K. If the value of $\frac{K}{P_B - P_A}$ is equal to $\frac{n\pi V_0^2 R^2}{9g}$. Find n.



2. A block of mass 'm' is given velocity towards another block of mass 6kg kept at rest on a frictionless horizontal floor some distance apart from a wall as shown in the figure. If all collisions are elastic, find minimum value of mass 'm' (in kg) so that blocks will collide only once.



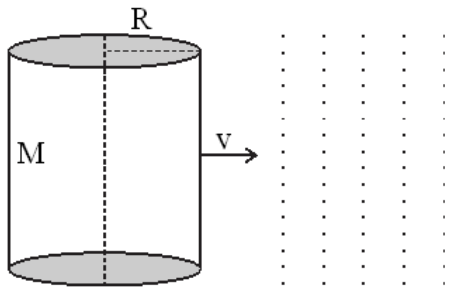
3. In a hydrogen like atom an electron is orbiting in an orbit having quantum number n. Its frequency of revolution is found to be 13.2×10^{15} Hz. Energy required to free the electron from the atom from the above orbit is 54.4 eV. In time 7 nano-seconds the electron jumps back to orbit having quantum number $\frac{n}{2}$.

If τ be the average torque acted on the electron during the above process, then find $\frac{\tau}{3} \times 10^{27}$ in N–m.

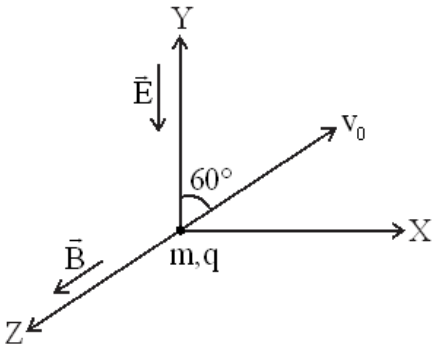
(Given $\frac{h}{\pi} = 2.1 \times 10^{-34}$ J–s).

Frequency of revolution of electron in the ground state of H-atom $f_0 = 6.6 \times 10^{15}$ Hz and ionisation energy of H-bond $E_0 = 13.6$ eV)

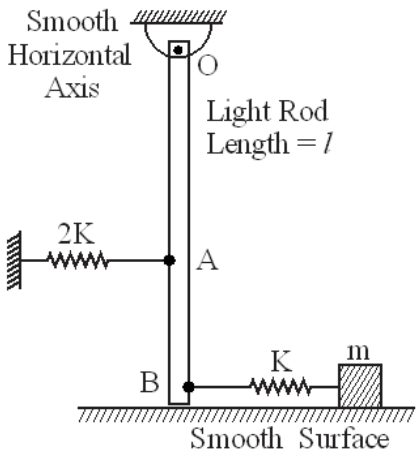
4. As shown in the figure a cylinder of mass M and radius R moves with constant speed v through a region of space that contains dust particles of mass m which are at rest. Number of particles in unit volume is n. The cylinder moves in a direction perpendicular to its axis. Assume that the particles do not interact with each other and that m < < M. All collisions are elastic and surface of the cylinder is smooth. The force per unit length of the cylinder required to maintain the speed v constant is $\frac{K\ nmRv^2}{3}$. Find the value of K.



5. A positively charged particle is given initial velocity v_0 in XY plane as shown in diagram. Initially the particle was at origin and the region contains uniform, mutually perpendicular electric and magnetic field. Let $\vec{E} = -E\hat{j}$ and $\vec{B} = B\hat{k}$. The minimum magnitude of the magnetic field so that the charge particle crosses the origin again during its subsequent motion is $B_{min} = \frac{n\pi E}{v_0}$. Find n



6. In the arrangement shown, two springs are attached to a light rod as shown. Length of O to A is $\frac{2\ell}{3}$. System is at equilibrium. The time period of oscillation of the block, if it is displaced slightly is found to be $T = 2\pi\sqrt{\frac{\alpha m}{8K}}$. Find the value of α.



PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. A metal exists in two allotropic forms, α and β, having a face centred cubic and hexagonal closest packed lattice respectively. Identify the correct statement.

(A) Coordination number is higher for the α-form.

(B) Density is higher for the β-form.

(C) Distance between closest octahedral voids is same for both the forms.

(D) Unit cell volume is same for both the forms.

2. Consider two Van der waal gases A and B. For gas A, $\lim_{P\to 0}\left(\frac{\partial Z}{\partial P}\right)_{T=300K} > 0$ and for gas B, $\lim_{P\to 0}\left(\frac{\partial Z}{\partial P}\right)_{T=300K} < 0$

Identify the incorrect statement.

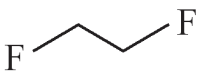
(A) Boyle temperature of gas A < 300 K

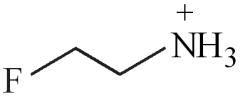
(B) Boyle temperature of gas B > 300K.

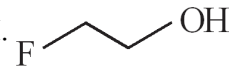
(C) Boyle temperature of both gases is 300 K

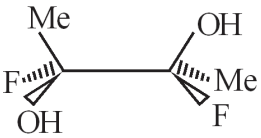
(D) Critical temperature of B is higher than that of A

3. In which of the following cases, the most stable conformer is achiral?

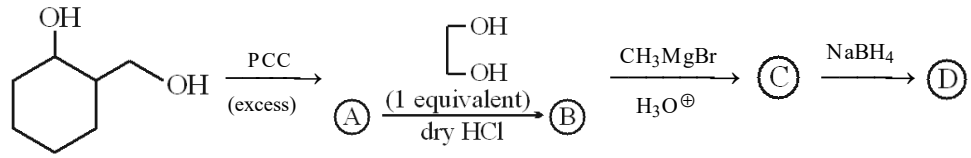
I. 

II. 

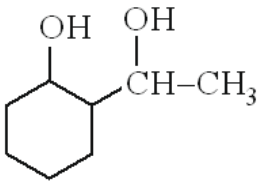
III. 

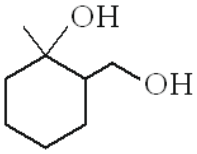
IV. 

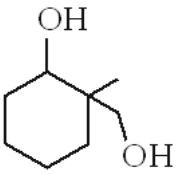
(A) Only IV (B) I and IV (C) I, II and III (D) In all the cases

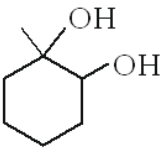
4. 

product 'D' is:

(A) 

(B) 

(C) 

(D) 

5. Select the incorrect statement.

(A) The less stable tautomeric form of H_3PO_3 has three active hydrogens.

(B) Both H_3PO_3 and H_3PO_2 disproportionate on heating to produce PH_3 and H_3PO_4.

(C) In P_4O_{10} all the O-P-O bond angles are identical whereas there are two different P-O bond lengths.

(D) H_3PO_3 and H_3AsO_3 have different basicity.

6. Which of the following statement is correct regarding extraction of aluminium from bauxite ore:

(A) In Hall's process CO_2 is formed at the anode.

(B) Baeyer's process is employed when the bauxite ore is rich in silica.

(C) In Serpeck's process N_2 is produced due to hydrolysis of aluminium nitride.

(D) Bauxite ore undergoes roasting before the leaching process.

SECTION-I(ii) : (Maximum Marks: 24)

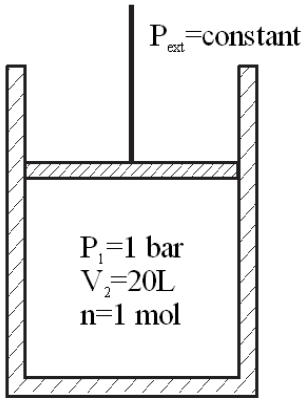
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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7. 1 mole of He $\left(C_{v,m} = \frac{3R}{2}\right)$ at 1 bar is compressed adiabatically against a constant external pressure, P_{ext} from an initial volume of 20 L. Assuming He to show ideal gas behaviour, identify the correct statement(s) for the process. (Take R = 2 cal $mol^{-1}K^{-1}$, ln3 = 1.1, ln5 = 1.6)



(A) The lowest possible final volume of the gas after this process is 8L.

(B) If P_{ext} = 3 atm, then $\Delta S = 0.8$ $calK^{-1}$

(C) ΔS_{sys} is +ve and $\Delta S_{Surroundings} = 0$ for all values of P_{ext}

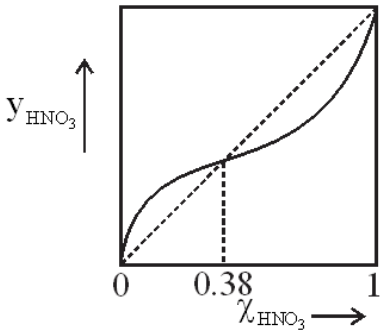
(D) If T_1, V_1, are the initial temperature and volume of the system, and T_2, V_2 are the final temperature and volume respectively, then $\ln\left(\frac{T_2}{T_1}\right) = \frac{R}{C_v}\ln\left(\frac{V_1}{V_2}\right)$

8. An azeotrop of HNO_3 and H_2O is formed with 38 mole percent of HNO_3. Identify the correct plot(s).

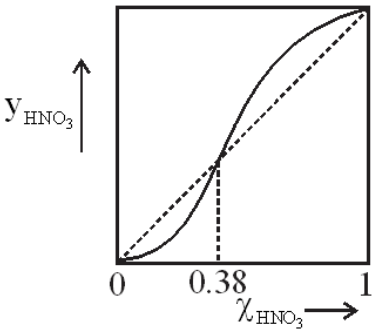
y → vapour phase mole fraction.

χ → liquid phase mole fraction.

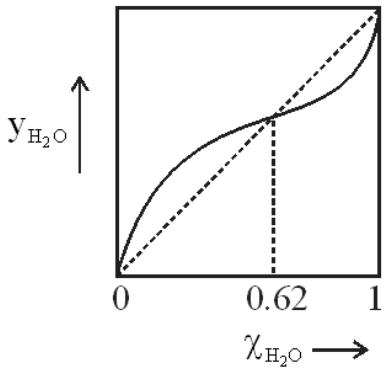
(A)



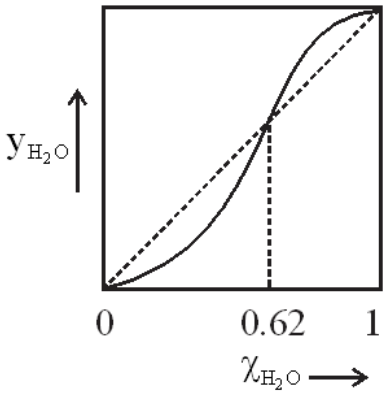
(B)



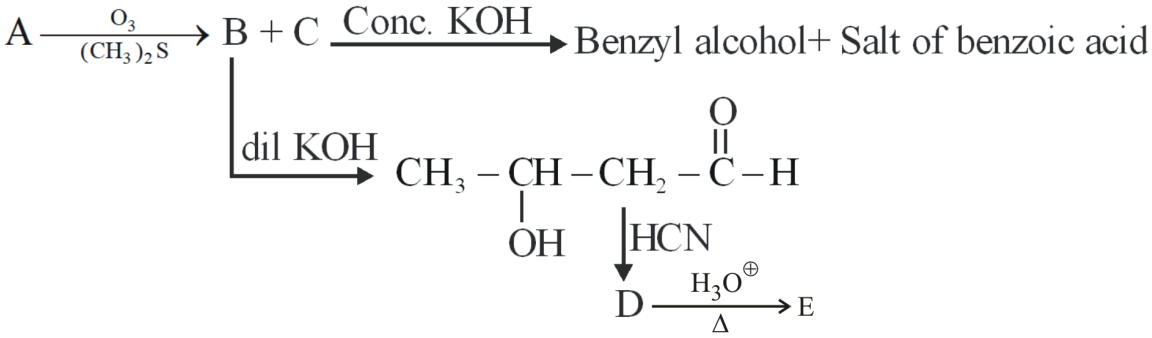
(C)



(D)



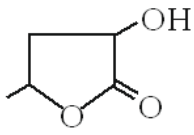
9.



Identify the correct statement(s):

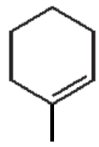
(A) Structure of 'E' is $CH_3-CH=CH-\overset{\overset{O}{\|}}{C}-H$.

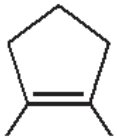
(B) Structure of 'B' and 'C' can be differentiated by Fehling solution.

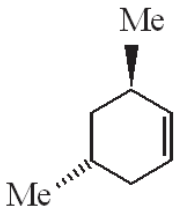
(C) Structure of 'E' is 

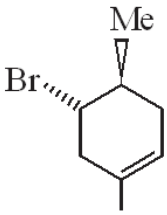
(D) Structure of 'A' is Ph – CH = CH – CH_3

10. Which of the following is/are not correctly matched?

(A)  $\xrightarrow{DCl}$ Number of stereoisomers of intermediate and products are identical.

(B)  $\xrightarrow[CCl_4]{Br_2}$ Neither product nor intermediate exhibits optical activity.

(C)  $\xrightarrow[Pd]{H_2}$ The chair form of the product gives non-superimposable structure after ring flipping.

(D)  $\xrightarrow[PhCOOOCOPh]{HBr}$ Product mixture upon fractional distillation gives 4 major fractions, out of which 3 are optically active.

11. Select the correct statement(s):

(A) $[Cr(NH_3)_5SO_4]Cl$ will produce a white precipitate with $AgNO_3$ solution but not with $BaSO_4$ solution.

(B) Octahedral complexes formed by coordinating strong field ligands with Fe^{+3} will be diamagnetic in nature.

(C) $[Ni(en)_3]Cl_2$, will show optical as well as geometrical isomerism.

(D) $K[PtCl_3C_2H_4]$ is an organometallic compound

12. A (is a compound of Ba) and B (is a sulphate) are white crystalline substances, highly soluble in water and decompose at higher temperatures. If an aqueous solution of A (slightly basic) is added to an aqueous solution of B (slightly acidic) a white precipitate C forms. The filtrate is neutral and gives a brown colour solution with an excess of acidified KI solution. When boiled, the filtrate evaporates without the appearance of any residue. White solid D can be prepared by the heating of A in the absence of air. The reaction of D with water gives a colourless solution. This solution, if kept in an open container, slowly precipitates a white solid E and leaves water. Upon prolonged exposure to moist air at room temperature, solid D is transformed into E as well. E can be converted back into D but decomposition above 1400°C is required for this purpose. The reaction of B and D in water forms the precipitate C and is accompanied by a characteristic odour due to formation of a gas. A gives white precipitate with $AgNO_3$. Select the correct option(s):

(A) Compound A is on heating gives neutral, acidic and basic oxide

(B) Compound B is $(NH_4)_2SO_4$.

(C) Compound E is $BaCO_3$.

(D) I_2 will precipitate as brown coloured.

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

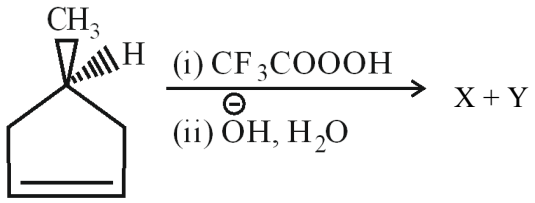
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

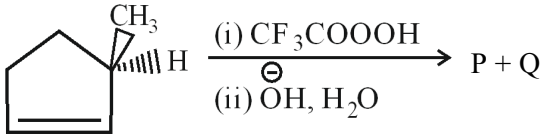
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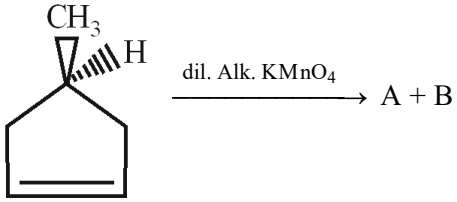
1. Hydrogen peroxide in aqueous solution decomposes by a first order reaction to produce water and oxygen. The rate constant for this decomposition is $7.50 \times 10^{-4}\ s^{-1}$. What quantity of heat (in J) is initially liberated per second from 2.0 L of solution that is 0.02M H_2O_2? Given $\Delta_f H°$ of $H_2O_2(l)$ = –187.0 kJ/mol and $\Delta_f H°$ H_2O (l) = –287.0 kJ/mol.

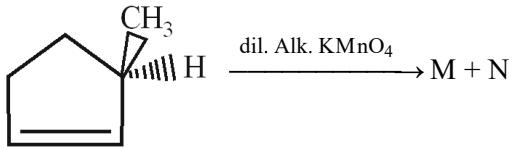
2. Calculate the pH of solution formed by addition of 1 mole each of HCl, CH_3COONa and HCOONa in enough water to make 1 L solution. The pK_a of HCOOH and CH_3COOH are 4 and 5, respectively.

3. In the given reactions:-

(i) 

(ii) 

(iii) 

(iv) 

Number of correct statement(s) among the following are :-

(1) X and Y are enantiomers

(2) M and N are enantiomers

(3) A and B are diastereomers

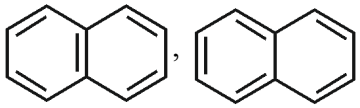
(4) (X and Y) are identical with (A and B)

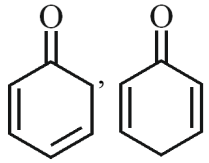
(5) (P and Q) are identical with (M and N)

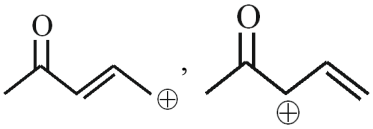
(6) M and N are geometrical isomers

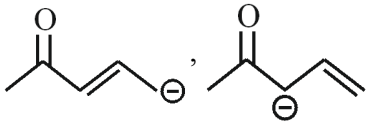
4. In how many of the below options, the first structure is more stable than the second.

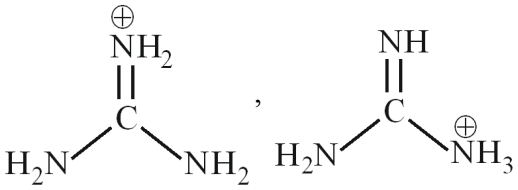
(1) 

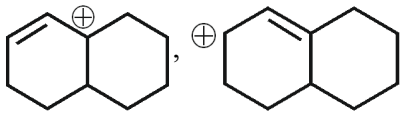
(2) 

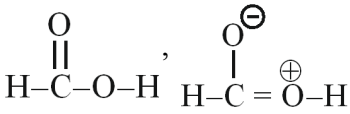
(3) 

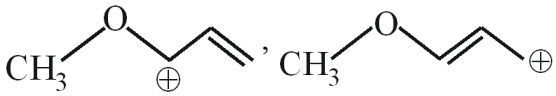
(4) 

(5) 

(6) 

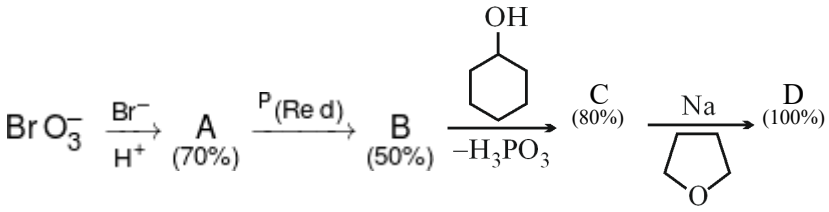
(7) 

(8) 

(9) 

5. If the reaction sequence given below is carried out with 4 mol of BrO_3^- and just required amount of Br^- and H^+, the amount of the product D formed (in g) is_____.

The yields of A, B, C, D are given in the parentheses.



6. The bond angle H-C-H in the formaldehyde molecule is approximately 120°; the dipole moments for C-H and C = O bonds are μ_{C-H} = 0.4 D and $\mu_{C=O}$ = 2.3 D, respectively. The net dipole moment of the formaldehyde is $X \times 10^{-Y}$ Cm (in scientific notation). The value of X is_____.

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. If α and β are exactly two distinct real roots of the equation $(x^2 + 5x - b^2)(2x^2 - ax - 6) = 0$ where $a, b \in R$, (Given that equation has only real roots) then

(A) $\alpha^2 + \beta^2 = 30$

(B) $a^2 + b^2 = 105$

(C) $\alpha^2 + \beta^2 = 19$

(D) $a^2 + b^2 + \alpha^2 + \beta^2 = 134$

2. If $f(x) = x^4 - 3x^2 + 7x - 1$ and $g(x) = \cot^{-1}(2x^2) + \sin^{-1}\left(\frac{1}{1+x^2}\right)$ are two functions, then number of integers in the range of gof(x) is

(A) 1

(B) 2

(C) 3

(D) 4

3. $\lim\limits_{x\to 0^+} \dfrac{(1+\tan x)^{2\cot x} - e^{2\cos x}}{1 - \cos(\ell n(1+\sin\sqrt{x}))}$

(A) $2e^2$

(B) $2e$

(C) $-2e$

(D) $-2e^2$

4. $\int e^x \left[\frac{2\sin x}{(\sin x + \cos x)^2} - \frac{2\cos x}{(\sin x - \cos x)^2} \right] dx$

where 'C' is the constant of integration.

(A) $e^x \left[\frac{2\sin x}{\sin^2 x - \cos^2 x} \right] + C$

(B) $e^x \left[\frac{2\cos x}{\sin^2 x - \cos^2 x} \right] + C$

(C) $e^x \left(\frac{\sin^2 x - \cos^2 x}{2\sin x} \right) + C$

(D) $e^x \left(\frac{\sin^2 x - \cos^2 x}{2\cos x} \right) + C$

5. The value of $S = (\sin^3\theta + \cos^3\theta) + (\sin^3 3\theta + \cos^3 3\theta) + (\sin^3 5\theta + \cos^3 5\theta) + \ldots\ldots\ldots\ldots + (\sin^3 19\theta + \cos^3 19\theta)$ when $\theta = \frac{\pi}{12}$ is $\frac{-a}{\sqrt{b}}$ where a & b are realtively prime numbers, then $|b - 6a|$ equals

(A) 1

(B) 2

(C) 3

(D) 4

6. Consider $f(x) = x^3 + ax^2 + bx + c$ where $a, b, c \in \{1, 2, 3, 4, 5, 6\}$. The probability that $f(x) = 0$ has all real roots given that f(x) is non-decreasing function, is

(A) $\frac{1}{216}$

(B) $\frac{1}{96}$

(C) $\frac{1}{108}$

(D) $\frac{1}{36}$

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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7. The vertices of triangle ABC are A(2, 0, 2); B(−1, 1, 1) and C(1, −2, 4). The point D and E divide AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that perpendicular drawn from F to the plane containing ΔABC, meets the plane at the point of intersection (point P) of the line segment CD & BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}$ units then

(A) $\dfrac{\text{area of } \Delta ABC}{\text{area of } \Delta PBC} = \dfrac{7\sqrt{3}}{6}$

(B) Volume of tetrahedral ABCF = $\dfrac{7}{3}$ cubic units

(C) Equation of line AF is $\vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{j})$

(D) Equation of line AF is $\vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} - 2\hat{k})$

8. If chords of the ellipse $\dfrac{x^2}{a^2} + y^2 = 2$, passing through (a, −1) are bisected by the line x + y = 1, then

(A) least positive integral value of a is 2.

(B) least positive integral value of a is 1.

(C) sum of integral values of a in (−10, 5) is −8.

(D) sum of negative integral values of a in (−20, −9) is −145.

9. Let $f : R \longrightarrow R$ be a differentiable function satisfying $f(x + y) = f(x) + f(y) + (x + y)\, e^{x+y} - xe^{x} - ye^{y} + 4xy$ $\forall\ x, y \in R$ and $f'(0) = 1$, then which of the following option(s) is/are **correct**?

(A) $f(x) = e^{-x+3}$; $x > 0$ has exactly one solution

(B) $f(2) - f'(2) < 0$

(C) $3f(x) = \text{sgn}(-x^2 + 3x - 6)$ has exactly one solution in $x \in [2, 5]$

(D) $f(3) - 3e^3 = x^2 - 6x + 5$ has exactly two solution

10. $\int_0^{\pi} [\,\{x\} + [\cot x]\,]\ dx$ is greater than or equal to

[Note: $\{z\}$ & $[z]$ denotes fractional part function and greatest integer function of z.]

(A) 0

(B) 1

(C) $-\frac{\pi}{2}$

(D) $-\pi$

11. Let OABC is a rhombus where O is origin such that vertices A (z_1) and C (z_3) lie on the circle $|z| = 2$ and the vertex B (z_2) lie on the circle $|z| = 3$. Let $\text{Arg}\,(z_2 - z_1) = \frac{\pi}{3}$, then

(A) $z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i}\right)(1 + \sqrt{3}i)$

(B) $z_2 = \left(\frac{6 + 6\sqrt{3}i}{3 - \sqrt{7}i}\right)$

(C) $z_3 = (1 + \sqrt{3}i)$

(D) $z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i}\right)(1 - \sqrt{3}i)$

12. Let $f(x) = -x^3 + px^2 + qx + 6\,\text{sgn}\,(x^2 + 2x + 5)$ where $p, q \in R$ & sgn denotes signum function. If the largest possible interval in which $f'(x)$ is positive is $(-2, 1)$, then $(p + q)$ is less than or equal to

(A) 4

(B) 5

(C) 6

(D) 9

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. Let a plane P containing a line $\vec{r} = \left(\hat{i} + \hat{j} + \hat{k}\right) + \lambda\left(-\hat{i} + 3\hat{j} + 2\hat{k}\right)$ and perpendicular to $2x + 3y - z = 2$. If the area of projection triangle formed on plane P whose vertices are $A_r = 2r\hat{i} + 3(-1)^r\hat{j} + r^2\hat{k}$ where $r \in N$ & $1 \leqslant r \leqslant 3$ is $a\sqrt{b}$ where a, b are relatively prime numbers, then $\left(\frac{b}{a}\right)$ is equal to

2. In a book with page numbers 1 to n, four consecutive pages are torn off and the sum of the numbers on the remaining pages is 1596, then find the sum of possible values of n. (4 consecutive pages means page numbers m, m+1, m+2 & m+3, $m \in N$)

3. From a fixed point A, three normals are drawn to the parabola $y^2 = 4ax$ at the points P, Q and R. Two circles C_1 and C_2 be drawn on AP and AQ as diameter. If slope of the common chord of the circles C_1 and C_2 be m_1 and the slope of the tangent to the parabola at the point R be m_2, then $m_1 \times m_2$ is equal to

4. Let S be the set containing all matrices $A = [a_{ij}]_{3\times3}$ having entries from the set {0,1,2,4} such that sum of all principal diagonal entries of $AA^T = 17$ then number of matrices in S is

5. If $\int_0^1 \left((6x^4 - f(x)\right) f(x)\, dx = 1$, then the area of region enclosed by $y = f(x)$, $x = 0$ and $x = 2$.

6. Let $u = \sin^7 x$, $v = \cos x$. If $\frac{d^2}{dx^2}\left(\frac{d^3u}{dv^3}\right)\Big|_{x=\frac{\pi}{8}} = 2^{\lambda_1} \cdot 3^{\lambda_2} \cdot 5^{\lambda_3} \cdot 7^{\lambda_4}$, then the value of $(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)$, is

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



English

(1001CJA106216230187)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
08-05-2024

JEE(Main+Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS		
Atomic No.	**:**	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :		H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

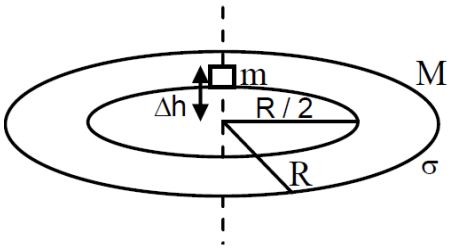
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. A small block of mass m lies above a thin disk of total mass M, constant surface density σ, and radius R. A hole of radius R/2 is cut into the center of the disk, allowing the mass to pass though. The mass starts at a position along the central axis of the disk but is displaced a height Δh from the plane of the disk, where Δh << R. There are no other external forces at work on this system; i.e., the disk and block are out in deep space. Assume m << M. For the first order in Δh the block undergoes simple harmonic motion with respect to the centre of the plane of the disk, compute the period of oscillation of block.



(A) $T = \sqrt{\frac{2\pi R}{G\sigma}}$

(B) $T = \sqrt{\frac{\pi R}{G\sigma}}$

(C) $T = \sqrt{\frac{\pi R}{2G\sigma}}$

(D) None of these

2. Consider a metal can placed coaxially inside a long solenoid. The metal can be modelled as a thin cylindrical shell of radius r. The magnetic field due to the solenoid varies with time t as kt where k is a positive constant. The resistivity of the can is ρ and its breaking stress is σ. Ignore the self-inductance of the can and any currents induced in the flat surfaces of the can. Assume that the magnetic field produced by the induced currents is negligible. Compute the time at which the can breaks.

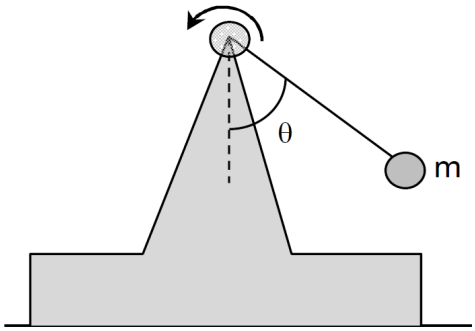
(A) $t = \frac{\sigma\rho}{r^2K^2}$

(B) $t = \frac{2\sigma\rho}{r^2K^2}$

(C) $t = \frac{\sigma\rho}{2r^2K^2}$

(D) $t = \frac{\sigma\rho}{3r^2K^2}$

3. A motor is installed at the top of a pole rigidly fixed on a platform. A light rod of length r = 1 m is rigidly attached to the motor shaft at its one end and at the other end a small ball of mass m is attached. The rod can rotated in a vertical plane with the help of the motor. Total mass of the platform, the pole and the motor is η = 4 times the mass of the ball. The motor rotates the rod at a constant angular velocity. The platform is placed on a horizontal surface, where coefficient of friction is $\mu = 1\sqrt{3}$. At which minimum angular velocity ω_0 of the rod, will the platform start sliding?



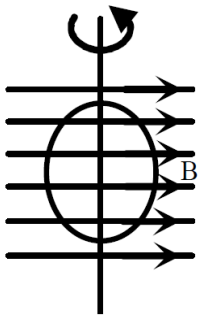
(A) $\omega_0 = 2\text{rad / s}$

(B) $\omega_0 = 3\text{rad / s}$

(C) $\omega_0 = 5\text{rad / s}$

(D) None of these

4. A ring of mean radius r and cross-sectional area A is made of a perfectly conducting wire. Inductance L of the ring is so small that inertia of free electrons cannot be neglected in the current building process. The free electron density in the conductor is n, mass of an electron is m and modulus of charge on an electron is e. Initially the ring is placed in a uniform magnetic field with its plane parallel to induction vector $\vec{B}$ of the field as shown in the figure. Find the current ring after it is turned through an angle 90°.



(A) $i = \dfrac{B\pi r^2}{L + \frac{m\times\pi r}{ne^2A}}$

(B) $i = \dfrac{B\pi r}{L + \frac{m\times\pi r}{ne^2A}}$

(C) $i = \dfrac{B\pi r^2}{L + \frac{m\times 2\pi r}{ne^2A}}$

(D) $i = \dfrac{B\times 2\pi r}{L + \frac{m\times\pi r^2}{ne^2A}}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

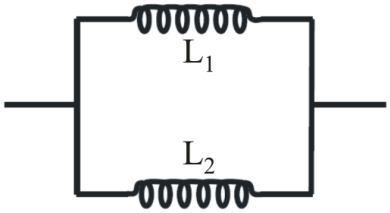
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

5.



For the given figure consider their mutual inductance is M then

(A) $Leq = \dfrac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$ When mutual inductance is not opposing

(B) $Leq = \dfrac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is not opposing.

(C) $Leq = \dfrac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is opposing.

(D) $Leq = \dfrac{L_1L_2 - M^2}{2L_1 + L_2 + 2M}$ When mutual inductance is opposing.

6. In a photoelectric effect experiment, a point source of light of power 40W emits mono-energetic photons of wavelength λ_1 that can just exit photoelectrons from an isolated metallic sphere of radius 1 cm, placed at a distance of 1 km from the light source. Now, three other sources of wavelength λ_2 ,λ_3 and λ_4 which are emitting the same number of photons as that of λ_1 brought near the source of λ_1. Assume photo efficiency of 10^{-6} (Take $\lambda_1 = 4960$Å, $\lambda_2 = 4133.33$Å, $\lambda_3 = 5000$Å, $\lambda_4 = 7200$Å and hc = 12.400 eV.Å)

(A) Number of photoelectrons emitted from the sphere per second is 2.5×10^9

(B) The potential of the sphere when emission of photoelectron stops is 0.8 V.

(C) The potential of the sphere when emission of photoelectrons stops is 0.5 V.

(D) The time after which the emission of photoelectrons will stop is 1.39×10^{-3} sec.

7. A dog running with a constant speed v is chasing a cat that is running with a constant velocity $\vec{u}$. During the chase, the dog always heads towards the cat. At an instant, direction of motion of the dog makes angle θ with that of the cat and the distance between them is r.

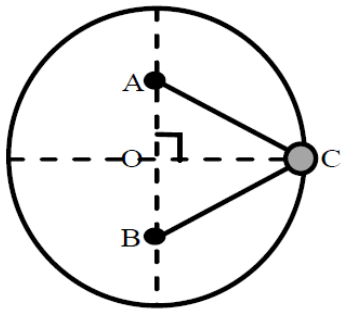
(A) The magnitude of acceleration of dog at this instant is $\frac{uv\sin\theta}{r}$.

(B) Radius of curvature of path followed by dog at this instant is $\frac{vr}{u\sin\theta}$.

(C) The magnitude of acceleration of dog at this instant is $\frac{uv\cos\theta}{r}$

(D) Radius of curvature of path followed by dog at this instant $\frac{ur}{v\cos\theta}$

8. A small bead of mass m can slide on a frictionless fixed ring of radius r. With the help of two identical strings of force constant k, the bead is connected to two nails A and B each on a diameter at a distance 0.5 r from centre O of the ring. Relaxed length of each string is negligible as compared to the radius of the ring. The bead is given a small velocity at point C. What can you predict about subsequent motion of the bead before any of the string strikes a nail? [ignore gravity]



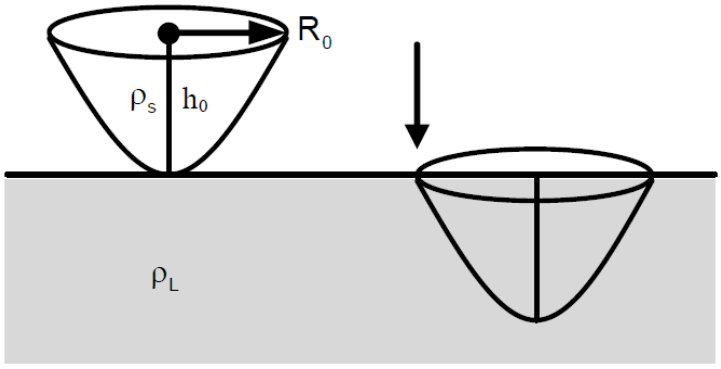
(A) It will keep moving with its initial speed.

(B) It will oscillate simple harmonically about the point C.

(C) Its angular momentum about centre O of the ring is conserved.

(D) Total elastic potential energy stored in both the strings is $1.25kr^2$.

9. A solid paraboloid of base radius R_0 and height h_0 is having uniform volume mass density of ρ_s is Inverted and placed just above the surface as shown, of a liquid having volume mass density of ρ_L. When paraboloid is released from rest it has been observed that when it becomes completely submerged in the liquid the paraboloid comes to the rest again. There exist uniform gravity of g. Assume that the liquid body is very large so that liquid level is not changing as the paraboloid enters into the liquid.



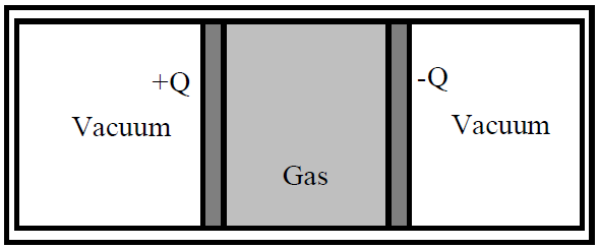
(A) The phenomenon is true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

(B) The phenomenon is not true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

(C) The ratio of $\frac{\rho_L}{\rho_s}$ is 2.

(D) The ratio of $\frac{\rho_L}{\rho_s}$ is 3.

10. A ideal gas having f degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of +Q and −Q. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased k times with the help of an external agent. Then which of the following statements are correct



(A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2+f}{2+f}$.

(B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2+f}{2+f}$.

(C) Work done by the agent in the process is $2(K^2-1)nRT_1$ where T_1 is initial temperature.

(D) Work done by the agent in the process is $\left(K^2-1\right)\frac{Q^2L_1}{A\varepsilon_0}$ where L_1 is initial separation between pistons.

SECTION-II (i) : (Maximum Marks: 12)

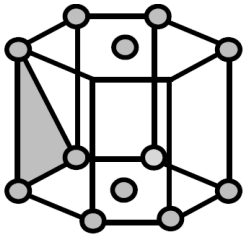
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

There is a hollow prism with hexagonal base of side L_0 and height H_0 as shown. There are identical non conducting uniformly charged solid spheres are placed at each corner and two identical spheres at the centre of flat hexagonal faces. The charges spheres have uniform density of d_0 and radius of R_0. What will be the flux passing through the shaded region and average charge density (ϕ) hexagonal prism. It is given that the flux passing through the hexagonal flat surface is k times the total flux passing through the prism.

(Take $\frac{R_0^3 d_0}{\varepsilon_0}$ = 1 Sl units & $\frac{d_0 R_0^3}{L_0^2 H_0}$ = 1 Sl units.)



1. The value of ϕ_s is_____

2. The value of d is_____

Paragraph for Questions 3 and 4

One side of a thin metal plate is illuminated by the sun. When the air temperature is T_0, the temperature of the illuminated side is T_1 and that of the opposite side is T_2. The temperature of each sides be T_1' & T_2', if another plate of double thickness is used. (Take T_0 = 300 K, T_1 = 330 K, T_2 = 320 K)

3. The value of T_1' (K) is _____

4. The value of T_2' (K) is _____

Paragraph for Questions 5 and 6

A boat is travelling in a river with a speed of 10 m/s along the stream flowing with a speed of 2 m/s. From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45 mm. Assume that attenuation of sound in water and air is negligible. The frequency detected by a receiver kept inside the river downstream is f_0 and the transmitter and the receiver are now pulled up into air. The air is blowing with a speed 5 m/sec in the direction opposite the river stream the frequency of the sound detected by the receiver is f_1. (Temperature of the air and water = 20°C; Density of river water = 10^3 kg/m^3; Bulk modulus of the water = 2.088×10^9 Pa; Gas constant R = 8.31J/mol –K ; Mean molecular mass of air = 28.8×10^{-3} kg/mol; Cp/Cv for air = 1.4)

Note: Boat velocity is with respect to ground & receiver is stationary w.r.t ground

5. The value of f_0 (KHZ) is_____

6. The value of f_1 (KHZ) is_____

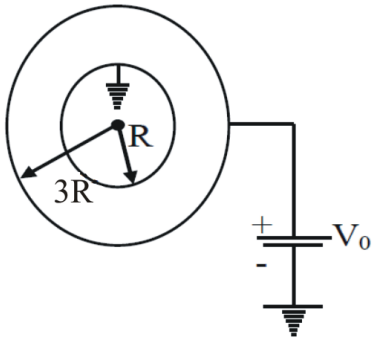
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

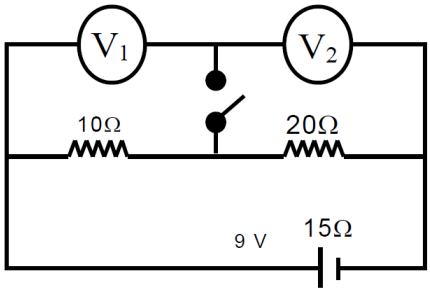
Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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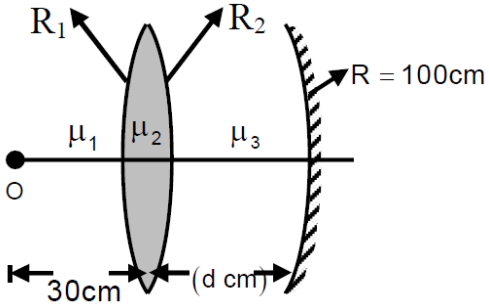
7. A conducting sphere with inner radius R and outer radius 3R has some alloys mixed in it due to which its resistivity changes with radial distance r according to the function $\rho_r = Kr$, where K is a +ve constant. Inner surface of the sphere is grounded while outer surface of sphere in maintained at potential V_0 with the help of a battery of emf V_0. Battery is ideal. The potential of the sphere at radial distance of 2R is $V_0\frac{\ell n\, a}{\ell n\, b}$. Find (a + b).



8. In the circuit shown in the figure the electromotive force of the battery is 9 V and its internal resistance is 15 Ω.The two identical voltmeters can be considered ideal. Let V_1 and V'_1 reading of 1st voltmeter when switch is open and closed respectively. Similarly, V_2 and V'_2 be the reading of 2nd voltmeter when switch is open and closed respectively. Then find the value of $\frac{V'_2 - V_2}{V_1 - V'_1}$.



9. The given thin lenses have $R_1 = R_2 = 10$ cm and $\mu_1 = 1$, $\mu_2 = 2$ and $\mu_3 = 4$. If final image formed at object and value of 'd' is x × 10 cm . Find 'x'.



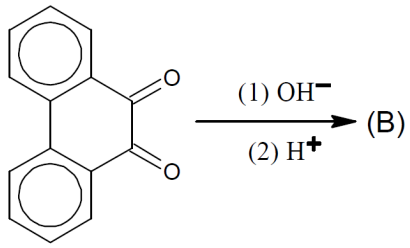
PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 12)

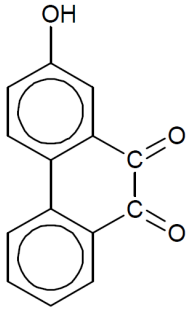
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

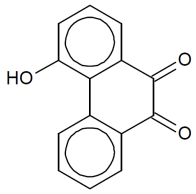
Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

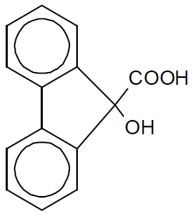
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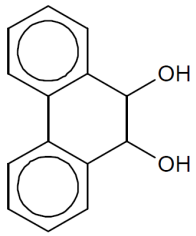
1. 

The compound (B) would be

(A) 

(B) 

(C) 

(D) 

2. How many methoxy group present in the molecule of a compound (A) would be present by the result of a Zeisel method. Treatment of compound (A) $C_{20}H_{11}O_4N$ with hot conc. HI gives CH_3I indicating presence of methoxy group. When 4.24 mg of a compound (A) is treated with hot conc. HI and CH_3I thus produced is digested with alc. $AgNO_3$, 11.62 mg of AgI is obtained.

(A) 1

(B) 2

(C) 3

(D) 4

3. The blue colour produced on adding H_2O_2 to acidified $K_2Cr_2O_7$ solution and shaking with ether is due to the formation of

(A) Cr_2O_3

(B) CrO_4^{-2}

(C) CrO_3

(D) CrO_5

4. A 0.001 molal solution of a complex $[MA_8]$ in water has the freezing point -0.0054^oC . Assuming 100% ionization of the complex salt and K_f for H_2O = 1.86 K-Kgmol^{-1}, write the correct representation of the complex.

(A) $[MA_8]$

(B) $[MA_7]A$

(C) $[MA_6]A_2$

(D) $[MA_5]A_3$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Cation(s) which form an ammine complex ion with excess of NH_3 is/are

(A) Ag^{+}

(B) Cu^{+2}

(C) Co^{+2}

(D) Al^{+3}

6. Which of the following statements regarding P_4O_{10} is correct ?

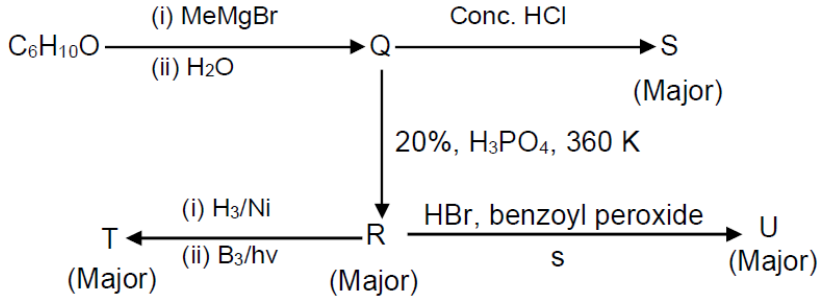
(A) It is an anhydride of metaphosphoric acid

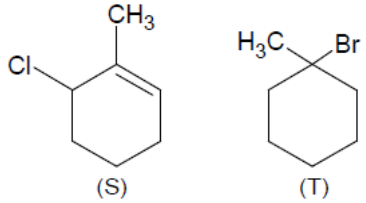
(B) It is a strong dehydrating agent

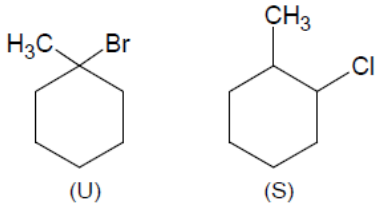
(C) In P_2O_5, each phosphorous is attached to three oxygen atoms through single bond and also with another oxygen atom through double bond.

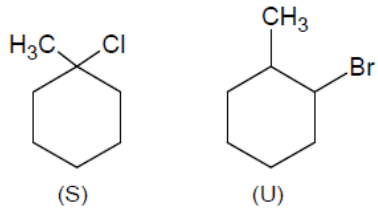
(D) The dehydrated product of HNO_3 obtained by treating it with P_4O_{10} is NO_2.

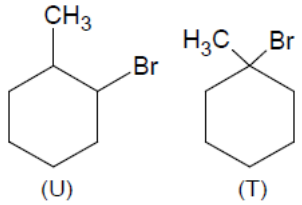
7. Choose the correct option(s) for the following set of reactions



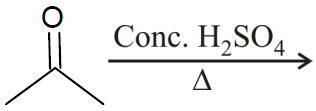
(A) 

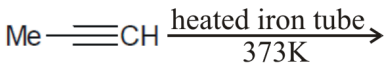
(B) 

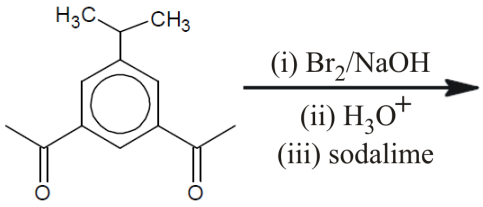
(C) 

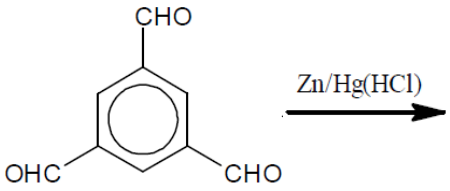
(D) 

8. The reaction(s) leading to the formation of 1, 3, 5 trimethyl benzene is/are

(A) 

(B) 

(C) 

(D) 

9. Which of the following statement(s) is/are correct?

(A) If temperature coefficient is greater than zero, cell reaction is endothermic.

(B) If temperature coefficient is less than zero, cell reaction is endothermic.

(C) If temperature coefficient is less than zero, cell reaction is exothermic.

(D) If E_{cell} is negative then ΔG is negative and cell reaction is spontaneous.

10. In which of the following pair of solution will the values of Vant Hoff factor be same? (Assuming same dissociation)

(A) 0.05 M K_4 $[Fe(CN)_6]$ and 0.10 M $FeSO_4$

(B) 0.10 M K_4 $[Fe(CN)_6]$ and 0.05 M $FeSO_4.(NH_4)_2$ $SO_4.6H_2O$

(C) 0.20 M NaCl and 0.10 $BaCl_2$

(D) 0.05 M $FeSO_4.(NH_4)_2$ $SO_4.6H_2O$ and 0.02 M $KCl.MgCl_2.6H_2O$

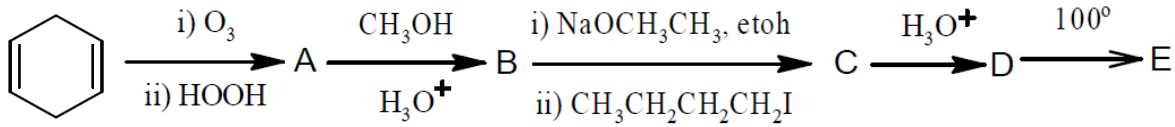
SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

For the following reaction scheme



1. The molecular weight of B is ……

2. The molecular weight of E is……

Paragraph for Questions 3 and 4

A 10^{-3} molal aqueous solution of a chromium complex having same number of ammonia molecule and Cl ions freezes at $-0.00558°C$. If all the ammonia molecules are acting as ligands and the percent of hydrogen in the complex is 4.6 (Given $K_f(H_2O) = 1.86K\ Kgmol^{-1}$) and assume 100% ionization self.

3. Find the number of Cl atom is complex is___.

4. The value of Vant Hoff factor is_____.

Paragraph for Questions 5 and 6

The isoelectric point (P_I) is the P_H at which the amino acid exists only as a dipolar ion with net charge zero.

Structure of lysine and aspartic acid are $C\overset{\Theta}{O}OCH\left(\overset{+}{N}H_3\right)(CH_2)_4NH_2$ and

$CO\overset{\Theta}{O}CH\left(\overset{+}{N}H_3\right)CH_2COOH$ respectively. The P_{Ka_1}, P_{Ka_2} and P_{Ka_3} of the dictation lysine are 2.18, 8.95 and 10.53 respectively. The P_{Ka_1}, P_{Ka_2} and P_{Ka_3} of the cation of aspartic acid are 1.88, 3.65 and 9.60 respectively.

5. Isoelectric point lysine is_____.

6. Isoelectric point of Aspartic acid is _____.

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

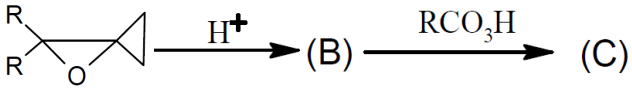
Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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7. Find the quantum numbers of the excited state of electrons (Let n_1 and n_2) in He^+ ion which on transition to ground state emit two photons of wavelength 30.4 nm and 108.5 nm respectively. Find the sum of $(n_1 + n_2)$ = ?

8. A compound AB forms NaCl type crystals and another compound XY forms CsCl type crystals. The formula. Mass of XY is three times that of AB but the cubic edge length of unit cell of AB crystals is twice that of XY. Calculate the ratio of density of XY to that of AB.

9. Size of ring in product formed C is ?



PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. Polynomial function f (x) satisfying the condition $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If f (10) = 1001, then f(20) is

(A) 7001 (B) 8001

(C) 8000 (D) None of these

2. Let $f(x) = e^{\left\{e^{|x|}\ \text{sgn}\ x\right\}}$ and $g(x) = e^{\left\{e^{|x|}\ \text{sgn}\ x\right\}}$, $x \in R$ where { } and [ ] denotes the fractional and integral part functions, respectively. Also h(x) = log(f(x)) + log (g(x)) then for real x, h(x) is

(A) An odd function

(B) An even function

(C) Neither odd nor an even function

(D) Both odd as well as even function

3. The length and foot of the perpendicular from the point (2, –1, 5) to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$ are

(A) $\sqrt{14}, (1, 2, -3)$

(B) $\sqrt{14}, (1, -2, 3)$

(C) $\sqrt{14}, (1, 2, 3)$

(D) None of these

4. If x = 9 is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangent is

(A) $9x^2 - 8y^2 + 18x - 9 = 0$

(B) $9x^2 - 8y^2 - 18x + 9 = 0$

(C) $9x^2 - 8y^2 - 18x - 9 = 0$

(D) $9x^2 - 8y^2 + 18x + 9 = 0$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Which of the following options are correct ?

(A) $\left[({}^nC_0+{}^nC_3+{}^nC_6+\ldots..) - \frac{1}{2}({}^nC_1+{}^nC_2+{}^nC_4+{}^nC_5+\ldots...)\right]^2 + \frac{3}{4}({}^nC_1-{}^nC_2+{}^nC_4-{}^nC_5+\ldots.)^2 = 1$

(B) If a and b are two positive numbers such that $a^5b^2 = 4$ then the maximum value of $\log_{2^{1/5}}(a^2) \cdot \log_{2^{1/2}}(b^2)$ is equal to 4

(C) Constant team in $((((((x-2)^2-2)^2-2)^2-2)^2 .....2)^2$ is equal to 2

(D) The coefficient of x^{24} in $\left(\frac{{}^{25}C_1}{{}^{25}C_0} - x\right)\left(x - 2^2\frac{{}^{25}C_2}{{}^{25}C_1}\right)\left(x - 3^2\frac{{}^{25}C_3}{{}^{25}C_2}\right) \times \left(x - 4^2\frac{{}^{25}C_4}{{}^{25}C_3}\right) \ldots\ldots$ $\left(x - 25^2\frac{{}^{25}C_{25}}{{}^{25}C_{24}}\right)$ is equal to 2925

6. $a_1, a_1, a_2, \ldots\ldots$ are distinct terms of an A.P. We call (p,q,r) an increasing triad if a_p, a_q, a_r are in G.P. where $p,q,r \in N$ such that $p < q < r$. If (5, 9, 16) is an increasing triad, then which of the following options is/are correct

(A) If a_1 is a multiple of 4 then every term of the A.P. is an integer

(B) (85,149,261) is an increasing triad

(C) If the common difference of the A.P. is $\frac{1}{4}$, then its first term is $\frac{1}{3}$

(D) ratio of $(4k + 1)$th term and $4k^{th}$ term can be 4

7. If $\int \frac{e^{x-1}}{(x^2 - 5x + 4)} 2x dx = AF(x-1) + BF(x-4) + C$ and $F(x) = \int \frac{e^x}{x} dx$, then

(A) $A = -2/3$

(B) $B = (4/3)e^3$

(C) $A = 2/3$

(D) $B = (8/3)e^3$

8. Let OA, OB, OC be coterminous edges of a cuboid. If l, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them respectively, then the value of $\frac{\left(\frac{1}{l^2} + \frac{1}{m^2} + \frac{1}{n^2}\right)}{\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)}$ is less than

(A) 2

(B) 3

(C) 4

(D) 5

9. Let the equation of a straight line L in complex form be $a\bar{z} + \bar{a}z + b = 0$, where a is a complex number and b is a real number, then

(A) the straight line $\frac{z-c}{a} + \frac{i(\bar{z}-\bar{c})}{a} = 0$ makes an angle of 45° with L and passes through a point c (where c is a complex number)

(B) the straight line $\frac{z-c}{a} = \frac{i(\bar{z}-\bar{c})}{a} = 0$ makes an angle of 45° with L and passes through a point c (where c is a complex number)

(C) the complex slope of the line L is $-\frac{a}{a}$

(D) the complex slope of the line L is $\frac{a}{a}$

10. The function $f(x) = x^{1/3}(x-1)$

(A) has two inflection points

(B) has one point of extremum

(C) is non – differentiable

(D) Range of f(x) is $[-3 \times 2^{-8/3}, \infty)$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Question Stem

Consider the locus of the complex number z in the Argand plane given by $\text{Re}(z) - 2 = |z - 7 + 2i|$. Let $P|z_1|$, $Q(z_2)$. Be two complex numbers satisfying the given locus and also satisfying $\arg\left(\frac{z_1 - (2 + \alpha i)}{z_2 - (2 + \alpha i)}\right) = \frac{\pi}{2}(\alpha \in R)$.

1. The minimum value of PR.QR where R represents the point (7, −2) is

2. $\arg\left(\frac{z_1 - (7 - 2i)}{z_1 - (7 - 2i)}\right)$ equals

Paragraph for Questions 3 and 4

Question Stem

T is the region of the plane $x + y + z = 1$ with $x, y, z > 0$. S is the set of points (a, b, c) in T such that just two of the following three inequalities hold : $a \leqslant \frac{1}{2}, b \leqslant \frac{1}{3}, c \leqslant \frac{1}{6}$.

3. Area of the region T is

4. Area of the region S is

Paragraph for Questions 5 and 6

Question Stem

Consider the line $\frac{x - 1}{2} = \frac{y}{-1} = \frac{z + 1}{2}$ and the point C(−1, 1, 2). Let D be the image of C in the line.

5. The distance C from the line is

6. The distance of the origin of the plane through the line and the point C is

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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7. A, B, C and D cut a pack of 52 cards successively in the order given. If the person who cuts a spade first receives Rs. 350 and if the expectation of A is λ then $\left[\frac{\lambda}{64}\right]$ is equal to (where [.] denotes greatest integer function)_____

8. Let $f(x) = \lim\limits_{x\to\infty} \frac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$. If f(x) is continuous for all $x \in R$, then the value of a + 8b is

9. A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that at least one of them occurs is 2/3. Find 8P(A) + 9P(B) if P(A) > P(B).

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____	_____
Signature of the Candidate	Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
08-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	D	B	C	C	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,D	A,C	A,B,C,D	B,D	A,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	2.00	5.00	8.00	2.00	17.00

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	C	A	B	C	A
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B,D	B,C,D	A,B,C	A,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	4.50	3.00	6.00	556.10 to 557.80	8.90 to 9.10

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	C	D	A	B	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	A,C,D	A,B,D	C,D	A,B,C	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.50	119.00	0.50	1206.00	19.20	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I(i)

1. Ans (C)

$$A_r = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$A_t = \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$\Rightarrow \frac{A_r}{A_t} = -\frac{1}{2} \text{ and } P = \frac{1}{2}\mu\upsilon\omega^2 A^2$$

$$= \frac{1}{2}\mu\sqrt{\frac{T}{\mu}}\upsilon\omega^2 A^2 \quad \left(\because \upsilon = \sqrt{\frac{T}{\mu}}\right)$$

$$P \propto \sqrt{\mu}\, A^2$$

$$\boxed{\frac{P_t}{P_r} = \sqrt{\frac{\mu_t}{\mu_r}}\left(\frac{A_t}{A_r}\right)^2}$$

$$= \sqrt{\frac{4\mu}{\mu}}(2)^2 = 8$$

2. Ans (D)

$$2R = \frac{3R}{2} + \frac{P\,dV}{dT}$$

$$\frac{P\,dV}{dT} = \frac{R}{2} \Rightarrow \frac{RT}{V}\frac{dV}{dT} = \frac{R}{2}$$

$$\Rightarrow \int \frac{dV}{V} = \frac{1}{2}\int \frac{dT}{T}$$

$$\ln V + \ln C = \frac{1}{2}\ln T \text{ (where C is constant)}$$

$$\ln CV = \frac{1}{2}\ln T \Rightarrow T = C^2V^2$$

Hence $T \propto V^2$

$\therefore$ Temperature becomes four times.

3. Ans (B)

$$I_0 = \frac{1}{4} \times (4M)\frac{a^2}{6}$$

$$= \frac{Ma^2}{6} = \frac{7 \times 144}{6} = 168\,kg\,cm^2$$

4. Ans (C)

Lens formula gives

$$\frac{1}{v} - \frac{1}{(-100)} = \frac{1}{50} \Rightarrow v = 100\,cm$$

The mirror forms an image 50 cm behind itself and the image is moving to right with a velocity of 20 m/s. The image formed by the mirror will act as moving object for the lens.

Again applying the lens formula gives

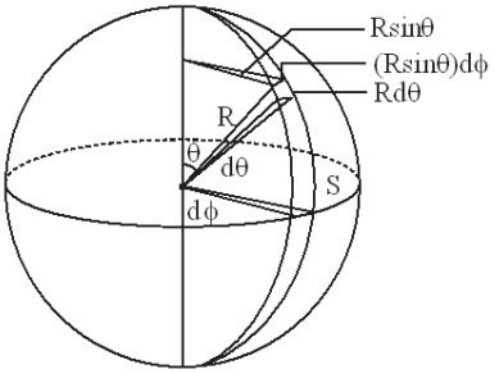
$$\frac{1}{v} - \frac{1}{(-200)} = \frac{1}{50} \Rightarrow v = \frac{200}{3}cm$$

For lens

$$\frac{dv}{dt} = \left(\frac{v}{u}\right)^2 \frac{du}{dt} = \left(\frac{200/3}{200}\right)^2 .20 = \frac{20}{9}cm/s$$

5. Ans (C)

The given charge distribution is equivalent to charge on a strip S on a uniformly charged spherical shell of radius R.



Since electric field in the sphere is zero, therefore net electric field due to any two diametrically opposite such strips on all points of the common diameter must be zero. It suggests that electric field of a single strip and hence electric field of the given semicircular arc anywhere on their diameter cannot have a component parallel to the diameter.

6. Ans (B)

$$\frac{dN}{dt} = t^2 - \lambda N$$

For $\frac{dN}{dt}$ to be minimum $\frac{d^2N}{dt^2} = 0$

$$\Rightarrow \frac{d^2N}{dt^2} = 2t - \lambda(t^2 - \lambda N) = 0$$

$$\text{Or } N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$$

PART-1 : PHYSICS

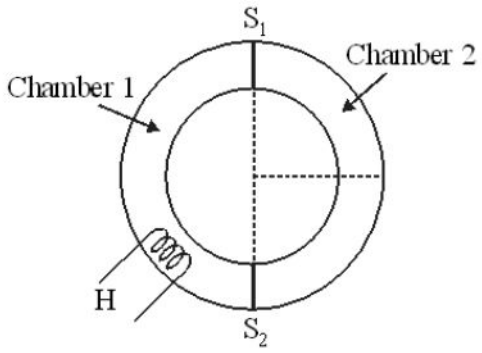
SECTION-I(ii)

7. Ans (B,D)

The process in chamber 2 is adiabatic.

Let final pressure in chamber 2 be P_2.

$$P_2\left(\frac{V_0}{2}\right)^{\gamma} = P_0V_0^{\gamma}$$



$$\Rightarrow P_2 = (2)^{3/2}P_0 \quad \left[\because \gamma = \frac{3}{2}\right]$$

$$= 2\sqrt{2}\,P_0$$

In equilibrium, pressure in both chambers must be same, hence final pressure in chamber 1 is

$$P_1 = P_2 = 2\sqrt{2}\,P_0$$

Work done on the gas in chamber 2 by gas in chamber 1 is

$$W = \frac{P_2V_2 - P_0V_0}{\gamma - 1} = \frac{2\sqrt{2}.P_0\frac{V_0}{2} - P_0V_0}{\frac{3}{2} - 1}$$

$$W = 2(\sqrt{2} - 1)\,P_0V_0$$

Change in internal energy of the gas in chamber 1

$$\Delta U = nC_v\Delta T = \frac{nR\Delta T}{\gamma - 1} = 2nR\Delta T$$

$$= 2[P_1V_1 - P_0V_0] = 2\left[2\sqrt{2}P_0.\frac{3}{2}V_0 - P_0V_0\right]$$

$$= 2(3\sqrt{2} - 1)\,P_0V_0$$

$$\therefore \Delta Q = \Delta U + W = 2(3\sqrt{2} - 1)\,P_0V_0 + 2(\sqrt{2} - 1)\,P_0V_0$$

$$= 4P_0V_0[2\sqrt{2} - 1]$$

8. Ans (A,C)

Loss in electrostatic potential energy

$U_i - U_f = \lambda E_0 \frac{L^2}{2}$

This is equal to gain in gravitational potential energy

$\lambda E_0 \frac{L^2}{2} = Mg\frac{L}{2} \Rightarrow E_0 = \frac{Mg}{\lambda L}$

The electric force on the rod is $F_e = \lambda L E_0$. Force can be considered at centre for writing the torque.

In equilibrium,

$\tau_{Fe} = \tau_{mg}$ [torque about hinge]

$\lambda L E_0 \frac{L}{2}\cos\theta = Mg\frac{L}{2}\sin\theta$

Putting $E_0 = \frac{Mg}{\lambda L}$ we get $\theta = 45°$

The rod has maximum KE at this position.

9. Ans (A,B,C,D)

$mg\left(h + \frac{H}{2}\right) = [v\rho g]\frac{H}{2}$

$h = H$

$mg\left(h + \frac{H}{2} + \frac{H}{2}\right) = [v\rho g]\frac{H}{2} + [v2\rho g]\frac{H}{2}$

$h = \frac{7H}{2}$

$mg\left(h + \frac{H}{2}\right) - [v\rho g]\frac{H}{2} = K.E.$

$mg\left(\frac{5H}{2}\right) - \frac{3}{2}mgH = K.E.$

K.E. = mgH.

10. Ans (B,D)

$W_g + W_{fr} = \Delta K$

W_g and ΔK same for all cases so W_{fr} should also be same

$N_1 < N_2$ and $\ell_1 < \ell_2 \Rightarrow \mu_1 > \mu_2$

[N = Normal forces, ℓ = length of the path]

$N_2 > N_3$ and $\ell_2 = \ell_3 \Rightarrow \mu_2 < \mu_3$

11. Ans (A,C)

For r = 0

Req = 1.5 R

for $r = \infty$

Req = 2R.

12. Ans (A,C,D)

Consider the X and Y coordinate axes along AB and BC respectively.

Time taken to collide BC is given by

$0 = \frac{ev\sqrt{3}}{2} - \frac{g\sqrt{3}}{2}t$

$\Rightarrow t = \frac{ev}{g}$

In x-direction

$2h = \left(\frac{v}{2}\right)t + \frac{1}{2}\left(\frac{g}{2}\right)t^2$

or $e^2 + 2e - \frac{8gh}{v^2} = 0$

$\therefore\ e = \frac{1}{\sqrt{3}}$

(B) $t = \frac{eV}{g} = \sqrt{\frac{8}{g}}\,s$

(C) $v_x = \frac{v}{2} + \frac{g}{2}t = \frac{v}{2}(1+e)$

$= \sqrt{6g}\left(1 + \frac{1}{\sqrt{3}}\right)$ m/s

PART-1 : PHYSICS

SECTION-II

1. Ans (8.00)

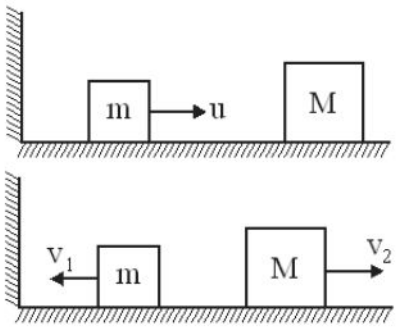
Liquid will be in pure translation

$P_B - P_A = 2R\rho g,$

$K = \frac{1}{2}mV_0^2 + \frac{1}{2}\left(mR^2 + \frac{2}{3}mR^2\right)\left(\frac{V_0}{R}\right)^2$

$\therefore\ \frac{K}{P_B - P_A} = \frac{8\pi R^2 V_0^2}{9g}$ $\therefore$ n = 8

2. Ans (2.00)

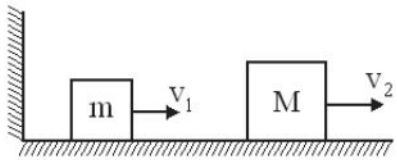


$$V_{cm} = \frac{mu}{m+M}$$

$$V_1 = -\frac{2mu}{m+M} + u$$

$$V_2 = \frac{2mu}{m+M}$$

For no collision, $V_2 \geqslant V_1$



$$\Rightarrow \frac{2mu}{m+M} \geqslant u - \frac{2mu}{m+M}$$

$$\Rightarrow \frac{4m}{m+M} \geqslant 1$$

$$\Rightarrow \boxed{3m \geqslant M}$$

3. Ans (5.00)

$$f = f_0\frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \text{ .....(1)}$$

$$E = E_0\frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \text{ .....(2)}$$

Solving we get n = 2, z = 4

$$L = mvr = \frac{nh}{2\pi}$$

$$\therefore \Delta L = \Delta n\frac{h}{2\pi}$$

Also, $\Delta L = \tau\ \Delta t$

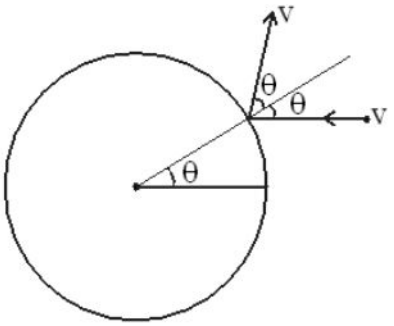
$$\therefore \tau\ \Delta t = \Delta n.\ \frac{h}{2\pi} \Rightarrow \tau = \frac{\Delta n}{\Delta t}\ \frac{h}{2\pi}$$

$$\Rightarrow \tau = \frac{1}{7\times10^{-9}} \times \frac{2.1\times10^{-34}}{2} = 15\times10^{-27}$$

$$\therefore \frac{\tau}{3}\times10^{27} = \frac{15\times10^{-27}}{3}\times10^{27} = 5$$

4. Ans (8.00)

In the frame of the cylinder the particles come in with speed v and then bounce off with same speed as shown.



Force

$$F = \int_{-\pi/2}^{\pi/2} (2mv\cos\theta)\cos\theta.\,(Rd\theta\cos\theta)\ell vn$$

$$\Rightarrow \frac{F}{\ell} = \frac{8}{3}mv^2nR \quad \therefore K = 8$$

5. Ans (2.00)

$$\bar{v} = \frac{v_0}{2}(\sqrt{3}\hat{i} + \hat{j})$$

For crossing origin,

$$K\frac{2\pi m}{qB} = \frac{2(v_0/2)}{(qE/m)}$$

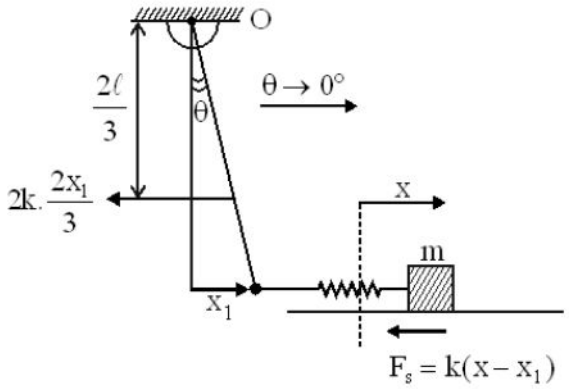
Where K has to be integer.

$$\Rightarrow B = \left(\frac{2\pi E}{v_0}\right)K$$

For minimum value K = 1

$$\Rightarrow B = \frac{2\pi E}{v_0}$$

6. Ans (17.00)



Light rod $\vec{\tau}_0 = 0$

$$2K.\frac{2x_1}{3}.\frac{2\ell}{3} = K(x - x_1).\ell$$

$$\Rightarrow \frac{17}{9}x_1 = x$$

Now for the block

$$K(x - x_1) = ma$$

$$\Rightarrow K.\left(x - \frac{9x}{17}\right) = ma$$

$$\Rightarrow -\frac{8.K}{17m}\vec{x} = \vec{a}$$

Time period is $T = 2\pi\sqrt{\frac{17m}{8K}}$

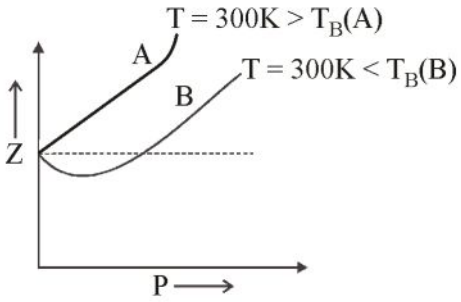
PART-2 : CHEMISTRY

SECTION-I(i)

1. Ans (C)

C.N, density and distance between closest octahedral voids (2r) is same for both hcp and fcc unit cells.

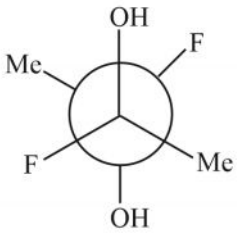
2. Ans (C)



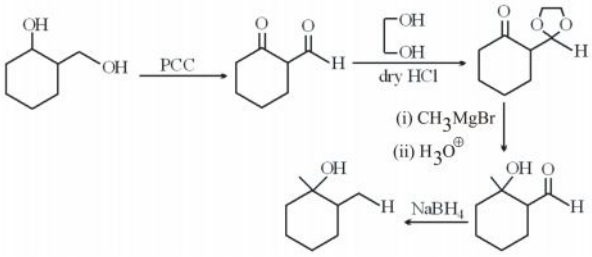
3. Ans (A)

I, II & III will have chiral gauche conformer as the most stable conformer.

IV- For the meso compound the most stable conformer is



4. Ans (B)



5. Ans (C)

Theory Based

6. Ans (A)

Theory Based.

PART-2 : CHEMISTRY

SECTION-I(ii)

7. Ans (A,B,C)

(a) $\Delta U = W$

$\Rightarrow \frac{P_2V_2 - P_1V_1}{\gamma - 1} = -P_{ext}(V_2 - V_1)$

$\Rightarrow \frac{P_2V_2 - 1 \times 20}{\frac{5}{3} - 1} = -P_2(V_2 - 20)$

$\Rightarrow 3(P_2V_2 - 20) = 2(-P_2V_2 + 20P_2)$

$\Rightarrow 5P_2V_2 - 40P_2 = 60$

$\Rightarrow P_2V_2 - 8P_2 = 12$

$\Rightarrow V_2 = \frac{12}{P_2} + 8$

As $P_2 \to \infty$, $V_2 \to 8$ (lowest possible value)

(b) $V_2 = \frac{12}{3} + 8 = 12L$

$\Delta S = nC_v \ln\left(\frac{P_2}{P_1}\right) + nC_P \ln\left(\frac{V_2}{V_1}\right)$

$= 1 \times \frac{3R}{2}\ln\left(\frac{3}{1}\right) + 1 \times \frac{5R}{2}\ln\left(\frac{12}{20}\right)$

$= \left[3\ln 3 + 5 \ln\left(\frac{3}{5}\right)\right]$ Cal K^{-1}

$= 0.8$ Cal K^{-1}

(c) For adiabatic process,

$\Delta S_{Surr} = 0$

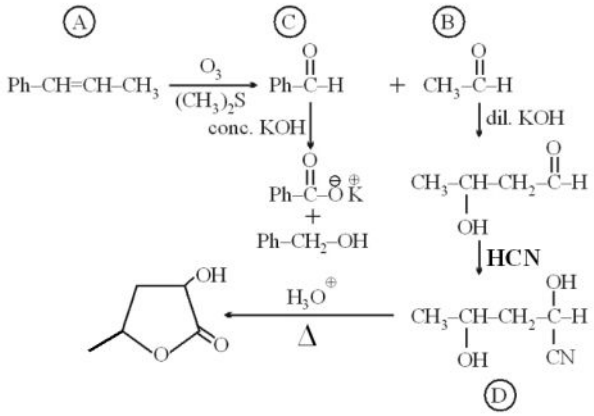
$\Delta S_{sys} = +ve$

(d) $TV^{\gamma-1}$ or $TV^{(R/CV)}$ is only valid for reversible processes.

8. Ans (B,D)

HNO_3 and H_2O form maximum boiling azeotrope and upon distillation, the distillate moves farther away from azeotrope composition, i.e. $y < \chi$ for $\chi < \chi_{azeotrope}$

9. Ans (B,C,D)



11. Ans (A,D)

(A) Only Cl is ionisable.

(B) Fe^{+3} has odd number of electrons($3d^5$), so paramagnetic.

(C) Only shows optical isomerism.

(D) As C_2H_4 bonds to Pt it's an organometallic compound.

12. Ans (A,B,C)

$Ba(NO_2)_2 = BaO + NO\uparrow + NO_2\uparrow$

(A) (D)

$(NH_4)_2SO_4 = NH_4HSO_4 + NH_3$

(B)

NH_4HSO_4 boils without further decomposition at 490°C.

$Ba^{2+} + SO_4^{2-} = BaSO_4$

(C)

$2NO_2^- + 2I^- + 4H^+ = 2NO + I_2 + 2H_2O$

$I_2 + I^- \rightleftharpoons I_3^-$ (Brown solution)

$NH_4^+ + NO_2^- = N_2 + 2\ H_2O$

$BaO + H_2O = Ba^{2+} + 2\ OH^-$

(D)

$Ba^{2+} + 2\ OH^- + CO_2 = BaCO_3\downarrow + H_2O$

(E)

$BaBO_3 \xrightarrow{\Delta} BaO + CO_2\uparrow$

(E) (D)

PART-2 : CHEMISTRY

SECTION-II

1. Ans (3.00)

$H_2O_2(aq) \rightarrow H_2O(\ell) + \frac{1}{2}O_2(g)$

$\Delta H_r = \Delta H^{\circ}_{f_{H_2O}} - \Delta H^{\circ}_{f_{H_2O_2}} = -287 - (-187)$

$= -100$ kJ/mol.

$\Rightarrow -\frac{d[H_2O_2]}{dt} = k[H_2O_2]$

$= 7.5 \times 10^{-4} \times 0.02 \text{ mol L}^{-1} \text{ s}^{-1}$.

$\therefore$ Heat evolved per second

$= (2 \times 7.5 \times 10^{-4} \times 0.02 \times 100 \times 10^3)\ \text{Js}^{-1} = 3\ \text{Js}^{-1}$.

2. Ans (4.50)

Initial concentration of $[HCOO^-]$, $[CH_3COO^-]$

and $[H^+] = 1M$

$HCOO^- + H^+ \rightleftharpoons HCOOH \quad K_1 = 10^4$

$(1-x) \quad (1-x) - \quad x$

$AcO^- + H^+ \rightleftharpoons AcOH \quad K_2 = 10^5$

$(1-y) \quad (1-x-y) \quad y$

$\because$ The equilibrium constants are very large, H^+

acts as the limiting reagent, and $1 - x - y \simeq 0$

$\Rightarrow x + y \simeq 1$ ...(i)

$K_1 = 10^4 = \frac{x}{(1-x)[H^+]}$

$\Rightarrow \frac{x}{1-x} = 10^4[H^+]$ ...(ii)

$K_2 = 10^5 = \frac{y}{(1-y)[H^+]}$

$\Rightarrow \frac{1-y}{y} = \frac{1}{10^5[H^+]}$

Also $y = 1 - x$

$\Rightarrow \frac{x}{1-x} = \frac{1}{10^5[H^+]}$ ...(iii)

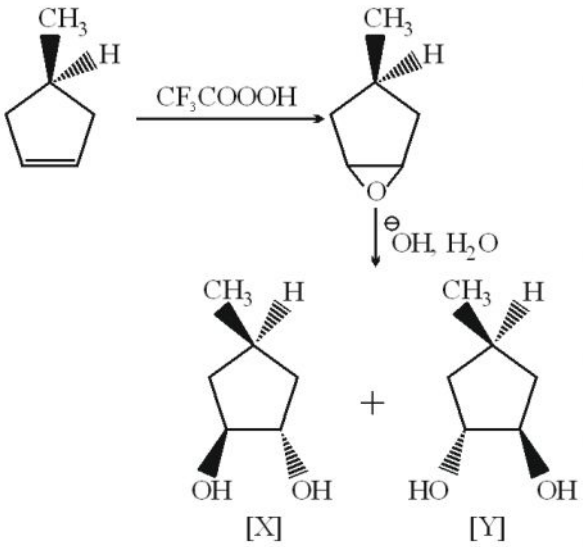
(ii) = (iii) $\Rightarrow 10^4[H^+] = \frac{1}{10^5[H^+]}$

$\Rightarrow [H^+]^2 = 10^{-9}$

$\Rightarrow$ pH = 4.5

3. Ans (3.00)

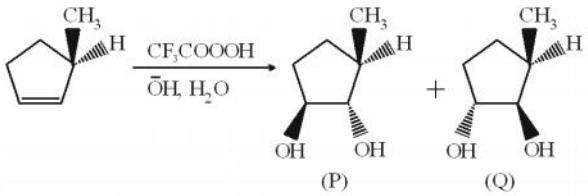
(i)



[X] [Y]

X and Y are enantiomer

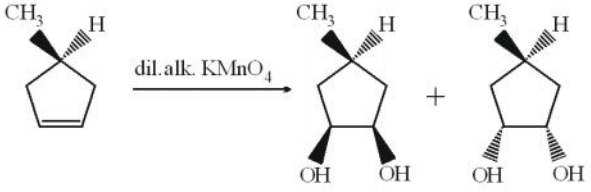
(ii)



(P) (Q)

P and Q are distereomer

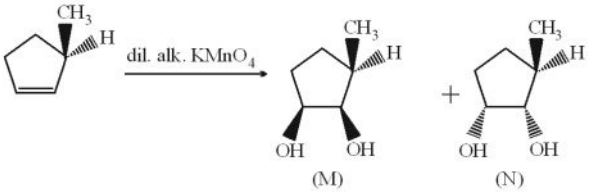
(iii)



(A) Meso (B) Meso

A and B are distereomer

(iv)

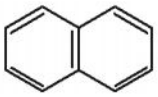


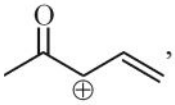
(M) (N)

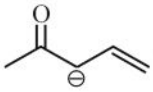
Correct statements are (1), (3) & (6)

4. Ans (6.00)

(1), (3) extended conjugation stabilizes more than cross conjugation

(2)  Both rings are aromatic

(4) In  , cation is destabilized by  group.

(5) In  anion is stabilized by group.

(6) Cation is not delocalised.

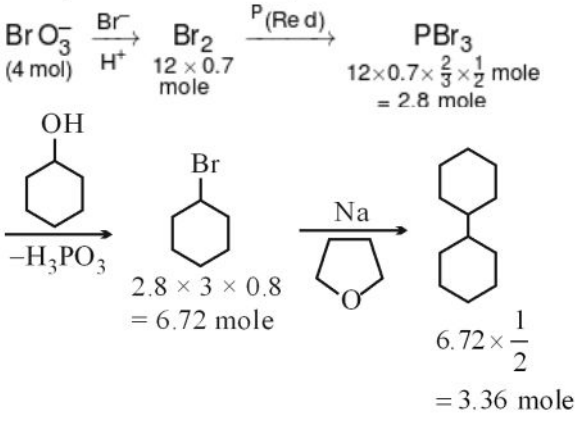
(7) 3° carbocation is more stable than 2° carbocation

(8) Uncharged resonating structure is more stable than charged.

(9) Cation is de-stabilised by -I effect.

Correct statements are 2,3,4,6,7, & 8.

5. Ans (556.10 to 557.80)



3.36 mole of D = 3.36 × 166 = 557.76

6. Ans (8.90 to 9.10)

dipole moment of HCHO = 0.4 + 2.3 = 2.7 D

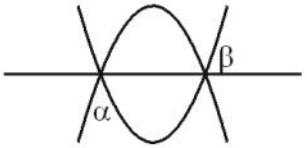
$2.7\ D = 2.7 \times 3.335 \times 10^{-30}\ C\ m = 9.00 \times 10^{-30}\ Cm$

PART-3 : MATHEMATICS

SECTION-I(i)

1. Ans (D)

$(x^2 + 5x - b^2)(-2x^2 + ax + 6) = 0$



$\Rightarrow$ both roots are in common $\frac{1}{-2} = \frac{5}{a} = \frac{-b^2}{6}$

$\Rightarrow a = -10,\ b^2 = 3$

Eqn. $x^2 + 5x - 3 = 0$

$\Rightarrow a + b = -5,\ \alpha\beta = -3$

2. Ans (C)

Range of f(x) contains $[0, \infty)$

$\therefore$ Range of gof(x) is $(0, \pi]$

$\Rightarrow$ Number of integers are 3.

3. Ans (D)

$$\lim_{x\to 0^+} \frac{((1+\tan x)^{\cot x} + e^{\cos x})((1+\tan x)^{\cot x} - e^{\cos x})}{\frac{1-\cos(\ell n(1+\sin\sqrt{x}))}{\ell n^2(1+\sin\sqrt{x})} \times \frac{\ell n^2(1+\sin\sqrt{x})}{(\sin\sqrt{x})^2} \times \frac{\sin^2\sqrt{x}}{(\sqrt{x})^2} \times x}$$

$$4e \lim_{x\to 0^+} \frac{(1+\tan x)^{\cot x} - e^{\cos x}}{x}$$

$$\Rightarrow 4e \lim_{x\to 0^+} \frac{e^{1-\frac{1}{2}\tan x + \frac{1}{3}\tan^2 x.....} - e^{\cos x}}{x}$$

$$= 4e \lim_{x\to 0^+} \frac{e^{\cos x}\left[e^{\left(1-\frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + ...\right) - \cos x} - 1\right]}{x}$$

$$= 4e^2 \lim_{x\to 0^+} \frac{\left(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + ...\right) - \cos x}{x}$$

$$= 4e^2\left(-\frac{1}{2}\right) = -2e^2$$

4. Ans (A)

$$\int \left(e^x \left(\frac{\sin x + \cos x}{(\sin x + \cos x)^2} - \frac{\cos x - \sin x}{(\sin x + \cos x)^2} \right) \right.$$

$$\left. + e^x \left(\frac{\sin x - \cos x}{(\sin x - \cos x)^2} - \frac{\cos x + \sin x}{(\sin x - \cos x)^2} \right) \right) dx$$

$$= e^x \left[\frac{\sin x + \cos x}{(\sin x + \cos x)^2} \right] + e^x \left(\frac{\sin x - \cos x}{(\sin x - \cos x)^2} \right] + C$$

$$= e^x \left[\frac{1}{\sin x + \cos x} + \frac{1}{\sin x - \cos x} \right]$$

$$= e^x \left[\frac{2 \sin x}{\sin^2 x - \cos^2 x} \right] + C$$

5. Ans (B)

$S = (\sin^3\theta + \sin^3 3\theta + \ldots\ldots\ldots\ldots + \sin^3 19\theta) +$

$(\cos^3\theta + \cos^3 3\theta + \ldots\ldots\ldots\ldots + \cos^3 19\theta)$

Using, $\sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$ and

$\cos^3\theta = \frac{\cos 3\theta + 3\cos\theta}{4}$

Let $S_1 = (\sin^3\theta + \sin^3 3\theta + \ldots\ldots\ldots\ldots + \sin^3 19\theta)$

$= \frac{3}{4}(\sin\theta + \sin 3\theta + \ldots\ldots\ldots\ldots \sin 19\theta)$

$-\frac{1}{4}(\sin 3\theta + \sin 9\theta + \ldots\ldots\ldots\ldots \sin 57\theta)$

$= \frac{3}{4}\left(\frac{\sin 10\theta}{\sin\theta} \cdot \sin 10\theta \right)$

$-\frac{1}{4}\left(\frac{\sin 30\theta}{\sin 3\theta} \cdot \sin 30\theta \right)$

For $\theta = 15$, $S_1 = \frac{3}{4}\left(\frac{1}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right)$

$-\frac{1}{4}(\sqrt{2}) = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}}$

and let $S_2 = (\cos^3\theta + \cos^3 3\theta + \ldots\ldots\ldots +$

$\cos^3 19\theta)$

$= \frac{3}{4}\left(\frac{\sin 10\theta}{\sin\theta} \cdot \cos 10\theta \right) + \frac{1}{4}\left(\frac{\sin 30\theta \cos 30\theta}{\sin 3\theta} \right)$

For $\theta = 15$,

$S_2 = \frac{3}{4}\left(\frac{-\sqrt{3}}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right) + \frac{1}{4}(0)$

$= \frac{-3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$

$S = S_1 + S_2 = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}} - \frac{3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$

$= \frac{3\sqrt{3}-1-9-3\sqrt{3}}{8\sqrt{2}} = \frac{-10}{8\sqrt{2}} = \frac{-5}{\sqrt{32}} \equiv \frac{-a}{\sqrt{b}}$

$\therefore a = 5, b = 32$

$\Rightarrow |b - 6a| = 2.$

6. Ans (B)

$\therefore$ f(x) is non-decreasing

$\Rightarrow f'(x) \geqslant 0$

$\Rightarrow 3x^2 + 2ax + b \geqslant 0 \Rightarrow D \leqslant 0$

$\Rightarrow 4a^2 - 4.3.b \leqslant 0 \Rightarrow a^2 \leqslant 3b$

$a = 1, b = 1, 2, 3, 4, 5, 6$

$a = 2, b = 2, 3, 4, 5, 6$

$a = 3, b = 3, 4, 5, 6$

$a = 4, b = 6$

$\Rightarrow$ a & b can vary in 16 ways

$\Rightarrow$ a, b, c can vary in $16 \times 6 = 96$ ways

$\therefore$ f(x) has all roots real, f(x) must be of the form $(x + \alpha)^3$

only possible case $(x+1)^3 \Rightarrow a = 3, b = 3, c = 1$

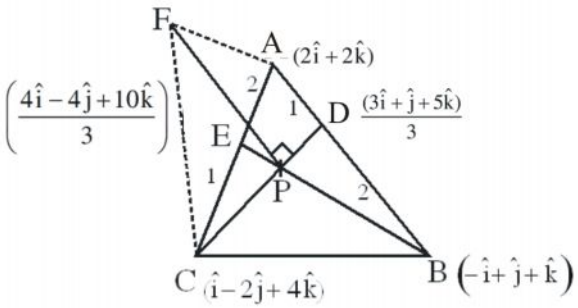
$\Rightarrow$ required probability $= \frac{1}{96}$

PART-3 : MATHEMATICS

SECTION-I(ii)

7. Ans (B,C,D)

CD; $\vec{r} = \left(\hat{i} - 2\hat{j} + 4\hat{k}\right) + \lambda\left(7\hat{j} - 7\hat{k}\right)$



BE; $\vec{r} = \left(-\hat{i} + \hat{j} + \hat{k}\right) + m\left(7\hat{i} - 7\hat{j} + 7\hat{k}\right)$

∵ Point of intersection of CD & BE is P

$\therefore P = \left(\hat{i} - \hat{j} + 3\hat{k}\right)$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & -1 \\ -1 & -2 & 2 \end{vmatrix}$$

$= \hat{i}(0) - \hat{j}(-7) + \hat{k}(7) = 7\hat{j} + 7\hat{k}$

$$\overrightarrow{PB} \times \overrightarrow{PC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & -2 \\ 0 & 1 & -1 \end{vmatrix}$$

$= (-2)\hat{j} + (-2)\hat{k} = -2\hat{j} - 2\hat{k}$

Area of tetrahedral ABCF

$= \frac{1}{3}(\text{Base Area}) \times \text{Height} = \frac{1}{3}\left(\frac{7\sqrt{2}}{2}\right) \times \sqrt{2} = \frac{7}{3}$

$\overrightarrow{PF} = \pm\sqrt{2}\left(\frac{\hat{j} + \hat{k}}{\sqrt{2}}\right) = (\hat{j} + \hat{k}) = \vec{F} - \vec{P}$

$\therefore \vec{F} = (\hat{i} - \hat{j} + 3\hat{k}) \pm (\hat{j} + \hat{k}) = \hat{i} + 4\hat{k};\ \hat{i} - 2\hat{j} + 2\hat{k}$

AF; $\vec{r} = 2\left(\hat{i} + \hat{k}\right) + \lambda\left(\hat{i} - 2\hat{k}\right);\ \vec{r} = 2\left(\hat{i} + \hat{k}\right) + \lambda\left(\hat{i} + 2\hat{j}\right)$

8. Ans (A,C,D)

Any point on the line x + y = 1 will be (t, 1 − t).

Equation of chord of the ellipse with this point as a midpoint

$$\frac{tx}{2a^2} + \frac{y(1-t)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

it passes through (a, −1) so that

$$\frac{at}{2a^2} + \frac{(t-1)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

$at + a^2(t-1) = t^2 + a^2(1-t)^2$

$(a^2+1)t^2 - (3a^2 + a)t + 2a^2 = 0$

Since we want two such chords

$D > 0$

$(3a^2 + a)^2 - 4(a^2+1) \cdot 2a^2 > 0$

$(3a+1)^2 - 8(a^2+1) > 0$

$a^2 + 6a - 7 > 0$

$(a+7)(a-1) > 0$

$a \in (-\infty, -7) \cup (1, \infty)$.

9. Ans (A,B,D)

$f(x+y) = f(x) + f(y) + (x+y) \cdot e^{x+y} - x \cdot e^x - y \cdot e^y + 4xy$

Put $x = y = 0 \Rightarrow f(0) = 0$

Differentiate partially w.r.t. x

$f'(x+y) = f'(x) + (x+y) \cdot e^{x+y} + e^{x+y} - x \cdot e^x - e^x + 4y$

Put $y \longrightarrow -x$

$f'(0) = f'(x) + 1 - x \cdot e^x - e^x - 4x$

$f'(x) = x \cdot e^x + e^x + 4x$

$f(x) = x \cdot e^x + 2x^2$

10. Ans (C,D)

$$\int_0^{\pi} [\{x\} + [\cot x]]\, dx = \int_0^{\pi} ([\{x\}] + [\cot x])\, dx$$

$$= \int_0^{\pi} [\cot\ x]\, dx$$

Now, $I = \int_0^{\pi} [\cot\ x]\, dx = \int_0^{\pi} [\cot\ (\pi - x)]\, dx$

$$2I = \int_0^{\pi} ([\cot\ x] + [-\cot x])\, dx$$

$$2I = \int_0^{\pi} (-1)\, dx = -\pi \Rightarrow I = -\frac{\pi}{2}$$

11. Ans (A,B,C)

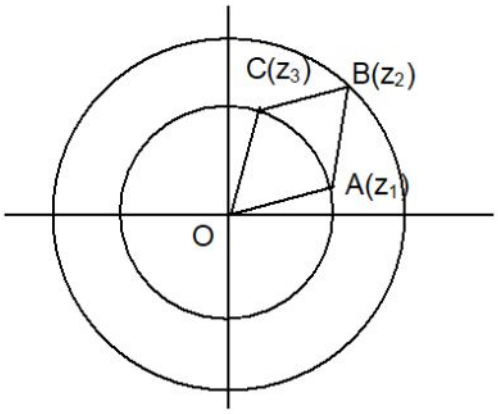
Given that $\text{Arg}\,(z_2 - z_1) = \frac{\pi}{3}$

since $\text{Arg}\, z_3 = \text{Arg}\,(z_2 - z_1) = \frac{\pi}{3}$

$\Rightarrow\ z_3 = |z_3|\, e^{i\pi/3}$

$\Rightarrow\ z_3 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$\Rightarrow\ z_3 = 2\left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) = (1 + \sqrt{3}i)$



The mid point of diagonals $= \frac{z_1 + z_3}{2} = \frac{z_2 + 0}{2}$

$\Rightarrow\quad z_1 + z_3 = z_2 \qquad \ldots (1)$

since $z_3 - z_1$ is perpendicular to z_2

$\Rightarrow \frac{z_3 - z_1}{z_2} = ki \Rightarrow z_3 - z_1$

$= ki\, z_2\ \ (k \in R) \quad \ldots (2)$

From (1) and (2) $z_1 = \left(\frac{1 - ki}{1 + ki}\right) z_3$

and $z_2 = \frac{2z_3}{1 + ki}$

$\Rightarrow |z_2| = \frac{2|z_3|}{|1 + ki|} \Rightarrow 3 = \frac{2 \times 2}{\sqrt{1 + k^2}}$

$\Rightarrow 1 + k^2 = \frac{16}{9}$

$\Rightarrow k = \pm\frac{\sqrt{7}}{3} \Rightarrow z_2 = \left(\frac{2}{1 \pm \frac{\sqrt{7}}{3}i}\right) \cdot (1 + \sqrt{3}i)$

$\Rightarrow z_2 = \frac{3(2 + 2\sqrt{3}i)}{3 + \sqrt{7}i}$ or $\frac{3(2 + 2\sqrt{3}i)}{3 - \sqrt{7}\,i}$ and

$z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i}\right)(1 + \sqrt{3}i)$

or $\left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i}\right)(1 + \sqrt{3}i)$

As two such rhombuses are possible, hence one

set of values is

$z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i}\right)(1 + \sqrt{3}i)$

$z_2 = \frac{6 + 6\sqrt{3}i}{3 + \sqrt{7}i}$ and $z_3 = (1 + \sqrt{3}i)$ and another

set of the values is

$z_1 = \left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i}\right)(1 + \sqrt{3}i)$,

$z_2 = \left(\frac{6 + 6\sqrt{3}i}{3 - \sqrt{7}i}\right)$ and $z_3 = (1 + \sqrt{3}i)$.

12. Ans (B,C,D)

$f'(x) = -3x^2 + 2px + q > 0\ \ \forall\, x \in (-2, 1)$

Sum of roots $\Rightarrow \frac{2p}{3} = -1 \Rightarrow p = \frac{-3}{2}$

and product of roots $\Rightarrow \frac{-q}{3} = -2 \Rightarrow q = 6$

$\therefore (p + q) = \frac{9}{2}$.

PART-3 : MATHEMATICS

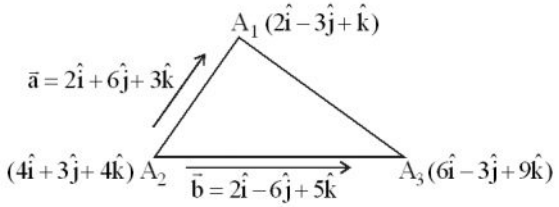
SECTION-II

1. Ans (9.50)

Normal vector of plane $\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & 2 \\ 2 & 3 & -1 \end{vmatrix}$

$= -9\hat{i} + 3\hat{j} - 9\hat{k}$

$\vec{a} \times \vec{b} = \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 3 \\ 2 & -6 & 5 \end{vmatrix} = 48\hat{i} - 4\hat{j} - 24\hat{k}$



Required area $= \frac{1}{2}\left|\vec{a} \times \vec{b}\right| \hat{n}_2 \cdot \hat{n}_1$

$= \frac{1}{2} 4\sqrt{181} \cdot \frac{228}{4\sqrt{181}\sqrt{171}} = 2\sqrt{19} \equiv a\sqrt{b}$

$\therefore \left(\frac{b}{a}\right) = 9.50.$

2. Ans (119.00)

Let page numbers k, k + 1, k + 2, k + 3 are torn off.

$\Rightarrow \frac{n(n+1)}{2} - (4k + 6) = 1596$

$\Rightarrow n^2 + n - 4(2k + 3) = 3192$

$\Rightarrow n^2 + n - 3192 = 4(2k + 3)$

$\Rightarrow (n + 57)(n - 56) = 4(2k + 3)$

if n = 59, k = 42

if n = 60, k = 57

Sum = 59 + 60 = 119

3. Ans (0.50)

Let $A \equiv (\alpha, \beta)$ the equation of the normal to the parabola at $(at^2, 2at)$ is

$y + tx = 2at + at^3$

which passes through (α, β)

So, $at^3 + (2a - \alpha)t - \beta = 0$

$\Rightarrow \quad t_1 + t_2 + t_3 = 0 \qquad \ldots(1)$

So, we have

$P\left(at_1^2, 2at_1\right)$, $Q \equiv \left(at_2^2, 2at_2\right)$ and $R \equiv \left(at_3^2, 2at_3\right)$

$C_1 \equiv$ Circle drawn on AP as diameter $= (x - \alpha)$

$(x - at_1^2) + (y - \beta)(y - 2at_1) = 0.$

$C_2 \equiv$ Circle drawn on AQ as diameter $= (x - \alpha)$

$(x - at_2^2) + (y - \beta)(y - 2at_2) = 0.$

Common chord $\equiv (t_1 + t_2)x + 2y - \alpha(t_1 + t_2) - 2\beta = 0 \quad (\text{as } t_1 \neq t_2)$

$\therefore \quad m_1 = -\left(\frac{t_1 + t_2}{2}\right) = \frac{t_3}{2} \qquad (\text{from (1)}$

Tangent to the parabola at R

$t_3 y = x + at_3^2 \Rightarrow m_2 = \frac{1}{t_3}$

so, $m_1 \times m_2 = \frac{1}{2}.$

4. Ans (1206.00)

Case-1: 4, 1, 0, 0, 0, 0, 0, 0, 0 $\longrightarrow \frac{9!}{7!} = 72$

Case-2: 2, 2, 2, 2, 1, 0, 0, 0, 0 $\longrightarrow \frac{9!}{4! \cdot 4!} = 630$

Case-3: 2, 2, 2, 1, 1, 1, 1, 1, 0 $\longrightarrow \frac{9!}{3! \cdot 5!} = 504$

$\therefore$ Number of matrices in S = 72 + 630 + 504 = 1206.

5. Ans (19.20)

$$\int_0^1 \left(6x^4 \cdot f(x) - f^2(x) - 9x^8 + 9x^8\right) dx = 1$$

$$\int_0^1 \left(9x^8 - \left(f(x) - 3x^4\right)^2\right) dx = 1$$

$$1 - \int_0^1 \left(f(x) - 3x^4\right)^2 dx = 1$$

$$\int_0^1 \left(f(x) - 3x^4\right)^2 dx = 0$$

$\therefore f(x) = 3x^4$

$$\therefore \text{ Required Area } = \int_0^2 (3x^4)\, dx = \frac{3}{5}x^5\Big|_0^2$$

$$= \frac{3}{5} \times (32 - 0) = 19.20.$$

6. Ans (5.00)

$u = \sin^7 x,\ v = \cos x$

$$\frac{du}{dv} = \frac{7\sin^6 x \cos x}{-\sin x} = -7\sin^5 x \cos x$$

$$\frac{d^2u}{dv^2} = 7\left[5\sin^3 x\cos^2 x - \sin^5 x\right]$$

$$\frac{d^3u}{dv^3} = -\frac{105}{4}\sin 4x$$

$$\text{Let } \frac{d^3u}{dv^3} = z \Rightarrow z = \frac{-105}{4}\sin 4x$$

$$\Rightarrow \frac{dz}{dx} = \frac{-105}{4}(4\cos 4x)$$

$$\Rightarrow \frac{d^2z}{dx^2} = -\,105\,(-4\sin 4x)$$

$$\left.\frac{d^2z}{dx^2}\right|_{x=\frac{\pi}{8}} = 105 \times 4$$

$$= 2^2 \cdot 3^1 \cdot 5^1 \cdot 7^1 \equiv 2^{\lambda_1} \cdot 3^{\lambda_2} \cdot 5^{\lambda_3} \cdot 7^{\lambda_4}$$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 5.$$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
08-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY — PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	A,C,D	A,B	A,C	A,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.35	4.83	333.33	316.67	100.69	103.04
SECTION-II (ii)	Q.	7	8	9			
	A.	5	1	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B,C	C	A,B,D	A,C	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	132.00	116.00	4.00	3.00	9.74	2.77
SECTION-II (ii)	Q.	7	8	9			
	A.	7	6	5			

PART-3 : MATHEMATICS

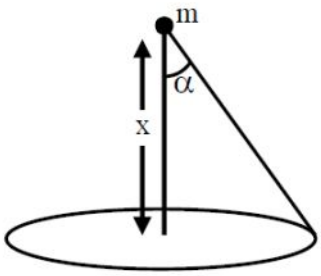
SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	A,B,C	A,D	B,C,D	A,B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	25.00	1.57	0.86	0.34	3.73	0.45
SECTION-II (ii)	Q.	7	8	9			
	A.	2	8	7			

HINT – SHEET

PART-1 : PHYSICS

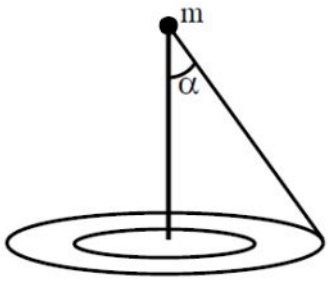
SECTION-I (i)

1. Ans (A)



$g = 2\pi G\sigma [1 - \cos \propto]$

$$g = 2\pi G\sigma \left[1 - \frac{x}{\sqrt{R^2 + x^2}}\right] = 2\pi G \left[1 - \frac{x}{R}\right]$$

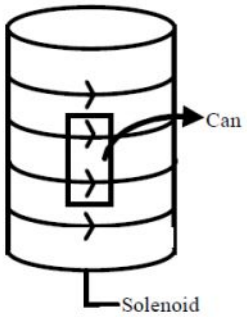


$$g = 2\pi G\sigma \left[\left(1 - \frac{x}{R}\right) - \left(1 - \frac{x}{R/2}\right)\right]$$

$$g = \frac{2\pi G\sigma}{R}x$$

$$T = \sqrt{\frac{2\pi R}{G\sigma}}$$

2. Ans (B)

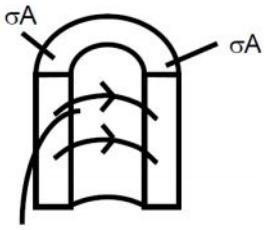


Emf is included in the can

$$Emf = \frac{dB}{dt}\pi r^2 = K\pi r^2$$

$$R = \frac{\rho \times 2\pi r}{A}$$

$$i = \frac{r}{2\rho}AK$$



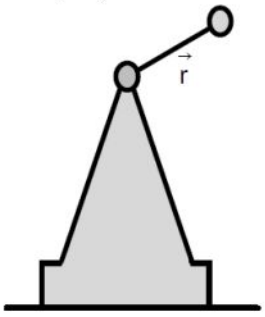
i2rB

From the diagram

$i \times 2rB = 2\sigma A$

$$t = \frac{2\sigma\rho}{r^2K^2}$$

3. Ans (C)



$\vec{F}$ on ball + $m\vec{g} = -mw^2\vec{r}$

$\vec{F}$ on platform = $m\vec{g} + mw^2\vec{r}$

$$m\omega^2 r = \frac{\mu 5mg}{\sqrt{1+\mu^2}} \quad \omega_0 = 5 \text{ rad/s}$$

4. **Ans (C)**

For perfect conductor R = 0 $E_{net} = 0$ in loop

ϕ_{net} = constant

$\phi_{net} = \phi_{self} + \phi_k + \phi_{ext}$

$= L_i + L_{ki} + \vec{B}.\vec{A} = L_{neti} + \vec{B}.\vec{A}$

$L_K = \dfrac{ml}{ne^2A}$ (Where m = mass of one electron, l = loop length)

As ϕ is constant $\phi_i = \phi_f$

Initially i = 0 & $\vec{B} \perp \vec{A}$

$\phi_{net} = 0$

Finally $\phi = -BA' + L_{net}i$

$i = \dfrac{B\pi r^2}{L_{net}} = \dfrac{B\pi r^2}{L + \frac{m\times 2\pi r}{ne^2A}}$

PART-1 : PHYSICS

SECTION-I (ii)

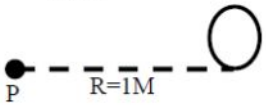
5. **Ans (A,C)**

$Leq = \dfrac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$ &

$Leq = \dfrac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$ (Opposing)

6. **Ans (A,C,D)**

$I = \dfrac{P}{4\pi R^2}$



Energy incident = $I\pi r^2$

No. of Photons incident/sec = $\dfrac{P \times \pi r^2}{4\pi R^2} \times \dfrac{\lambda}{hc}$

For λ_1 no. of photon incident/sec 2.5×10^{15}

Photo electric effect for λ_1 & λ_2 only.

$h\nu = \phi + K_{max} \Rightarrow K_{max} = 0$ (For λ_1)

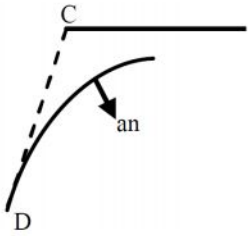
$E_2 = \dfrac{12400}{4133.33} = 3ev\phi = \dfrac{12400}{4960} = 2.5ev$

$V_s = 0.5$ volt

$\dfrac{KQ}{r} = V_s \Rightarrow Q_2 = ne$

7. **Ans (A,B)**

Acceleration of dog = Normal acceleration.



$\Delta A = \vec{A}_f - \vec{A}_i = \vec{A}_i + \Delta\vec{A}$

$a = \dfrac{dv}{dt} = \dfrac{vd\theta}{dt} = v\omega = \dfrac{v\sin\theta}{r}$

$R = \dfrac{v^2}{a_n} = \dfrac{vr}{u\sin\theta}$

8. **Ans (A,C)**

$\vec{F} = -2k\vec{r}$ toward 0.

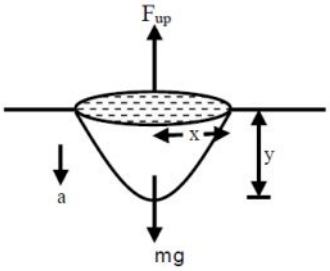
$\vec{\tau}_0 = 0$

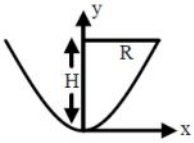
$P.t = \dfrac{1}{2}k\left(r^2 + \dfrac{r^2}{4}\right) = 1.25\,Kr^2$

9. **Ans (A,D)**

Let $Y = ax^2$ (A parabola)

$Y = \dfrac{H}{R^2}x^2$ (from given data)





$v = \dfrac{dy}{dt}$

$mg - \rho_L.\dfrac{1}{2}\pi x^2 y.g = m.\dfrac{dv}{dt}$

$\dfrac{dv}{dt} = g - \dfrac{\pi x^2 yg\rho_L}{2\rho_s\frac{\pi R^2 H}{2}}g \Rightarrow g - \dfrac{\rho_L}{\rho_S}\dfrac{x^2}{R^2} \times \dfrac{yg}{H}$

$v.\dfrac{dv}{dy} = g - \dfrac{\rho_L}{\rho_S}gy^2$

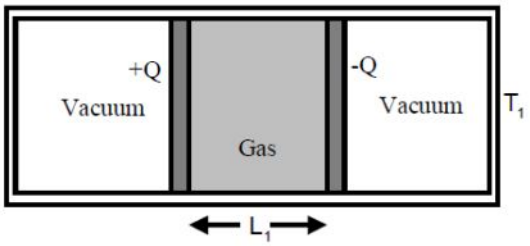
$\int_0^0 vdv = \int_0^H \left(g - \dfrac{\rho_L}{\rho_s}gy^2\right)dy$

$o = gH - \dfrac{g\rho_L}{3\rho_s}H$

$\dfrac{\rho_L}{\rho_s} = 3$

10. **Ans (B,C,D)**

Stage 1

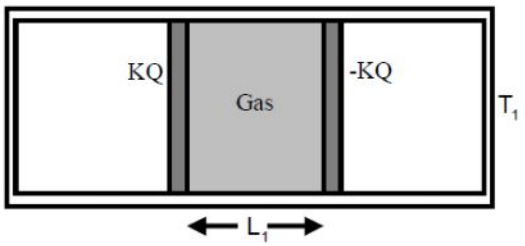


$P_1AL_1 = nRT_1$ $\quad \frac{Q_2}{2A\varepsilon_0} = P_1A = \frac{nRT_1}{L_1}$

(Force Balancing)

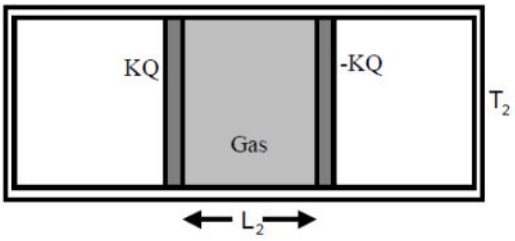
Stage 2

Charge is suddenly changed



System is not in equilibrium

Stage 3



L_2CL_1

Finally system in equilibrium

$P_2AL_2 = nRT_2$ $\quad \frac{K^2Q^2}{2A\varepsilon_0} = P_2A = \frac{nRT_2}{L_2}$

Energy of the system is conserved between stage 2 and 3.

$$\Rightarrow \frac{1}{2}\varepsilon_0 E_2^2 AL_1 + \frac{f}{2}nRT_1 = \frac{1}{2}\varepsilon_0 E_2^2 AL_2 + \frac{f}{2}nRT_2$$

On solving all the above equation

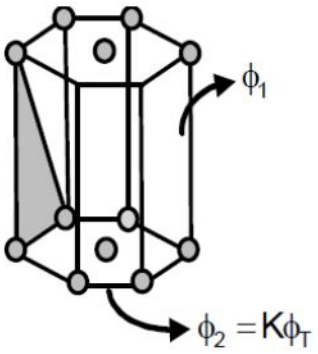
$$\frac{T_2}{T_1} = \frac{2K^2 + f}{2 + f}$$

$$w_{ext} = \left(K^2 - 1\right)\frac{Q^2 L_1}{2A\varepsilon_0} = \left(K^2 - 1\right)nRT_1$$

PART-1 : PHYSICS

SECTION-II (i)

2. **Ans (4.83)**

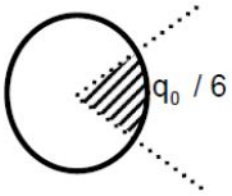


$6\phi_1 + 2\phi_2 = \phi_{total}$

$6\phi_1 + 2K\phi_{total} = \phi_{total}$

$$\phi_1 = \frac{(1-2K)\phi_{total}}{6} \Rightarrow \phi_{total} = \frac{q_{in}}{\varepsilon_0}$$

Top view



$$q_{in} = 12.\frac{q_0}{6} + 2\frac{q_0}{2}$$

$$\Rightarrow 3q_0 = q_{in} \Rightarrow 3d_0\frac{4}{3}\pi R_0^3$$

$$\phi_1 = \frac{(1-2k)}{6} \times \frac{4\pi {R_0}^3 d_0}{\varepsilon_0}$$

$$\phi_s = \frac{(1-2k)}{3}\frac{\pi {R_0}^3 d_0}{\varepsilon_0} = 0.35$$

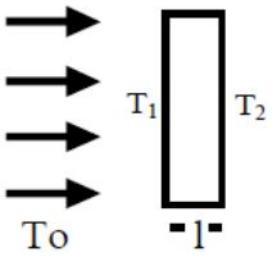
$$\text{Average charge density} = \frac{q_{in}}{V_{prism}}$$

$$d = \frac{8\pi}{3\sqrt{3}}\frac{d_0 {R_0}^3}{{L_0}^2 W_0} = 4.83$$

4. **Ans (316.67)**

Left face

Power from sun = P



$\frac{d\theta}{dt} \propto (T_1 - T_0)$ $\frac{d\theta}{dt} = \lambda (T_1 - T_0)$

$\lambda = 4\sigma e A T_0^3$

$P = \lambda (T_1 - T_0) + \frac{kA (T_1 - T_2)}{\ell}$ ...(i)

(For left surface)

$\frac{KA (T_1 - T_2)}{\ell} = \lambda (T_2 - T_0)$ ...(ii)

(For right surface)

When thickness is doubled

$P = \lambda (T'_1 - T_0) + \frac{kA (T'_1 - T'_2)}{2\ell}$ ... (iii)

$\frac{kA (T'_1 - T'_2)}{2\ell} = \lambda (T'_1 - T_0)$ ...(iv)

on solving

$T'_1 = \frac{2T_1^2 + T_1T_2 - T_0 (3T_1 - T_2) - T_2^2}{2 (T_1 - T_0)} = 333.33k.$

$T'_2 = \frac{T_1T_2 + T_0 (T_1 - 3T_2) - T_2^2}{2 (T_1 - T_0)} = 316.67k.$

6. **Ans (103.04)**

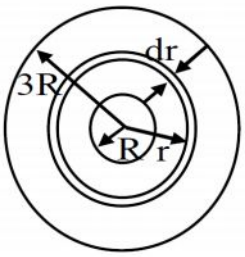
Use doppler effect.

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (5)**

$V_0 = I\ Req$



$R_{eq} = \int dR$

$R_{eq} = \int \frac{\rho_r dr}{4\pi r^2}$

$R_{eq} = \int_R^{3R} \frac{K dr}{4\pi r} \Rightarrow R_{eq} = \frac{K}{4\pi} \ell n(3)$

$R(r) = \frac{K}{4\pi} \ln \left(\frac{r}{R} \right)$

$V(r) = lR(r) = \frac{V_0}{R_{eq}} \times R(r)$

$V(2R) = V_0 \frac{\ell n2}{\ell n3}$

8. **Ans (1)**

When switch is open

Current in circuit

$I = \frac{9}{45} A = 0.2A$

$V_1 + V_2 = 6V$ and because in series same current passes and voltmeters are identical so

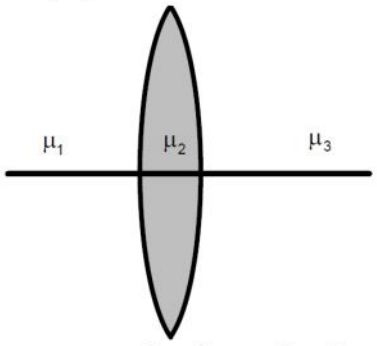
$V_1 = V_2 = 3V$

When switch closed

$V'_1 = 2V ; V'_2 = 4V$

$\frac{V'_2 - V_2}{V_1 - V'_1} = \frac{4-3}{3-2} = 1$

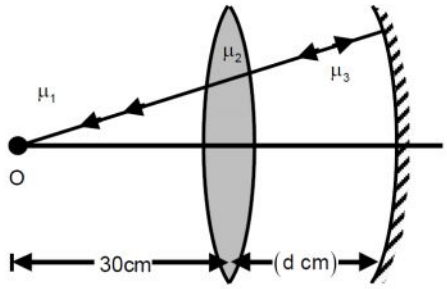
9. **Ans (7)**



$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$\frac{4}{v} - \frac{1}{-30} = \frac{2-1}{10} + \frac{4-2}{-10}$$

$v = -30$

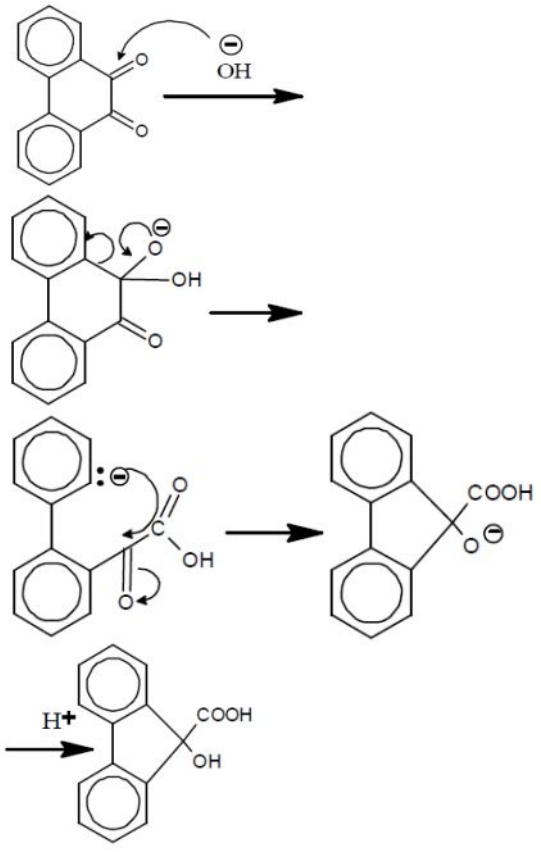


$d + 30 = 100$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (C)**



2. **Ans (D)**

$C_{20}H_{11}O_4N + HI \rightarrow CH_3I \xrightarrow{AgNO_3} AgI$

$\frac{1.24}{329} = .013$ $\qquad$ $\frac{11.62}{232} = 0.05$

No. of mole of AgI $\approx 4 \times$ mole of $C_{20}H_{11}ON$

3. **Ans (D)**

$K_2Cr_2O_7 + H_2O_2 + H_2SO_4 \xrightarrow{\text{ether}} \underset{\text{(Blue)}}{CrO_5}$

4. **Ans (C)**

$$\Delta T_f = iK_f \frac{w \times 1000}{M.W. \times W} = iK_f m$$

$.0054 = i\ 1.86 \times .001$

$$i = \frac{.0054}{1.86 \times .001} \approx 2.90$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C)**

$Ag^+ + 2NH_3 \rightarrow [Ag(NH_3)_2]^+$

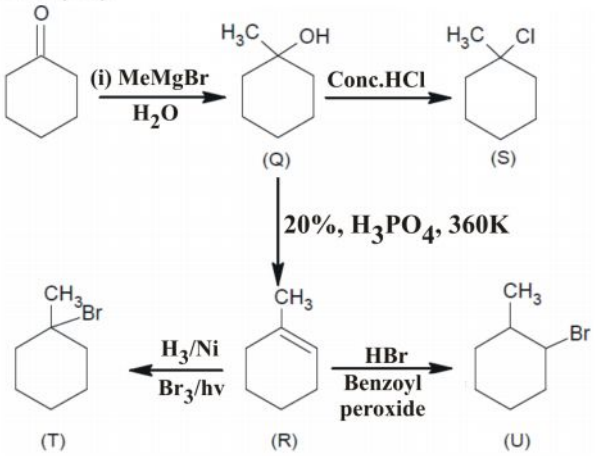
$Cu^{+2} + 4NH_3 \rightarrow [Cu(NH_3)_4]^{+2}$

$Co^{+2} + 6NH_3 \rightarrow [Co(NH_3)_6]^{+2}$

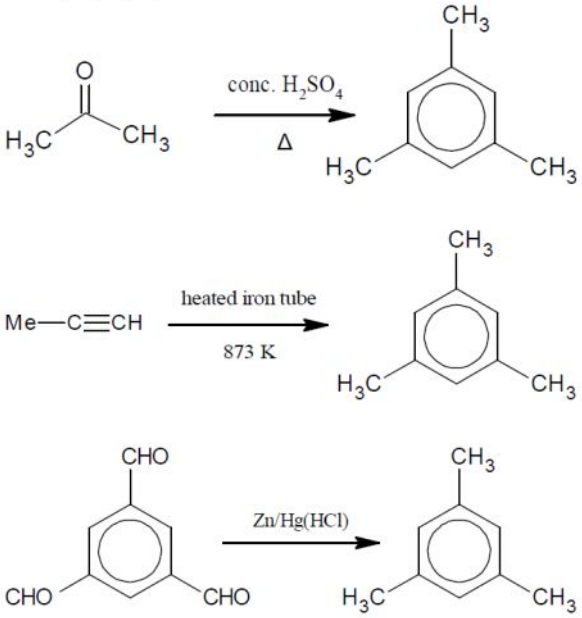
6. **Ans (A,B,C)**

HNO_3 on dehydration gives N_2O_5

7. **Ans (C)**



8. **Ans (A,B,D)**



9. **Ans (A,C)**

$\left(\frac{dE}{dT}\right) = \frac{E_2 - E_1}{T_2 - T_1}$, According to Nernst equation, on ↑ in temp. E.M.F. ↓ es.

10. **Ans (B,D)**

$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{-4}$

i = 5 No. of ions

$FeSO_4.(NH_4)_2 \quad SO_4.6H_2O \rightarrow Fe^{+2} + 2NH_4^+ + 2SO_4^{-2}$

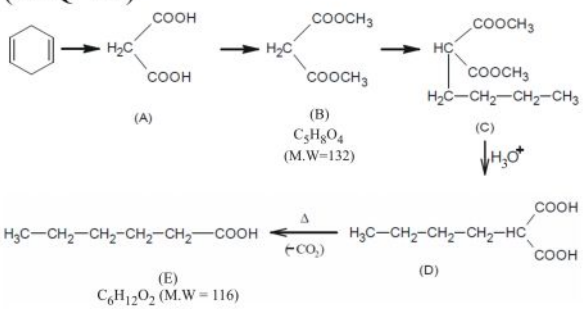
i = 5 = No. of ions

PART-2 : CHEMISTRY

SECTION-II (i)

2. **Ans (116.00)**

(for Q. 1-2)



4. **Ans (3.00)**

(for Q. 3-4)

Let the molecular formula of the complex is $Cr(NH_3)_xCl_x$

∴ % H is in the complex

$$\frac{3x}{52 + 17x + 35.5x} \times 100 = 4.6$$

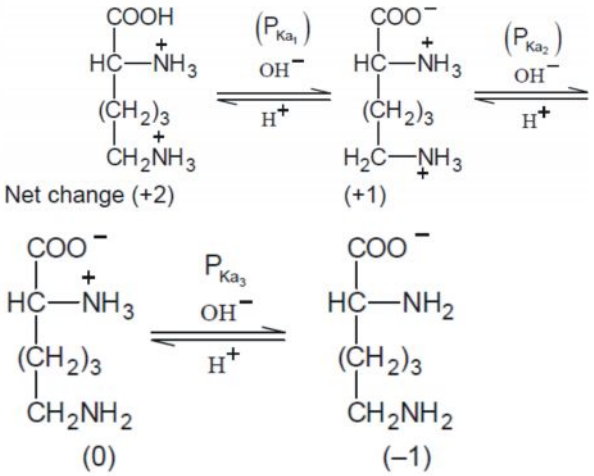
$x \approx 4.08 \approx 4$

Now, $\Delta T_f = i\, k_f m$

$$i = \frac{.00558}{1.86 \times 10^{-3}} \approx 3$$

5. **Ans (9.74)**

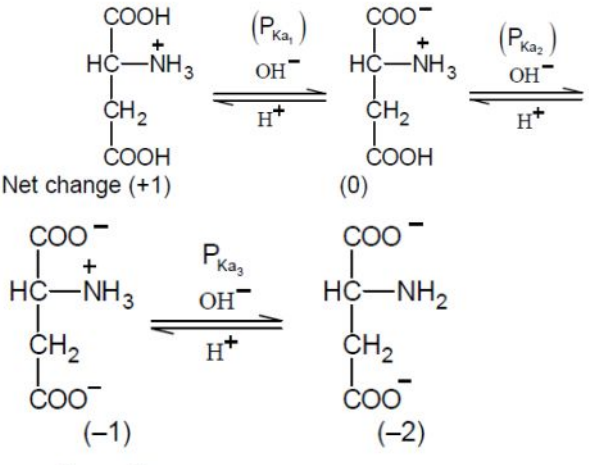
The dissociation of cationic form of lysine can be represented.



$$P_I = \frac{P_{Ka_2} + P_{Ka_3}}{2} = \frac{8.95 + 10.53}{2} = 9.74$$

6. **Ans (2.77)**

The dissociation of cationic form of Aspartic acid can be shown as



$$P_I = \frac{P_{Ka_2} + P_{Ka_2}}{2} = \frac{1.88 + 3.65}{2} = 2.77$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (7)**

$$\frac{1}{\lambda} = R_H(Z)^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

for $n_1 = 1$ $n_2 = ?$

$$\frac{1}{30.4\times10^{-9}} = 1.09678\times10^7\times4\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$$

$n_2 = 2$

for $n_1 = 2$ (first excited state), $n_2 = ?$

$$\frac{1}{108.5\times10^{-9}} = 1.09678\times10^7\times4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

$\boxed{n_2 = 5}$

sum of n_1 & $n_2 = 2 + 5 = 7$

8. **Ans (6)**

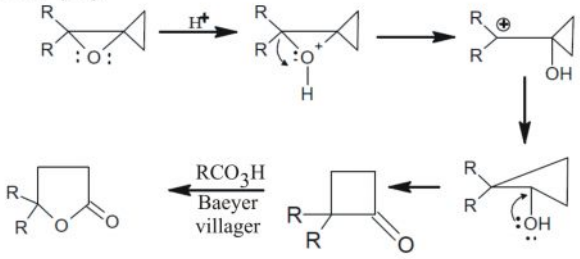
Density of AB $= \dfrac{4\times M_1}{N_{a_r}\times \ell_1^3}$ ....(i)

Let molar mass of AB be M_1 gm/mol and edge length of unit cell be ℓ_1(m).

Density of XY $= \dfrac{1.3M_1}{N_{a_r}\times\left(\frac{\ell_1}{2}\right)^3} = \dfrac{8\times3\times M_1}{N_{a_r}\times\ell_1^3}$ ....(ii)

(ii)/(i) = 6

9. **Ans (5)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

let $f(x) = a_nx^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \ldots + a_ax + a_0$

$$f\left(\frac{1}{x}\right) = \frac{a_n}{x^n} + \frac{a_{n-1}}{x^{n-1}} + a_{n-2}.\frac{1}{x^{n-2}} + \ldots\ldots + \frac{n}{x} + a_0$$

$$f(x).f\left(\frac{1}{x}\right) = a_0a_nx^n + (a_{n-1}a_0 + a_na_1)x^{n-1} + x^{n-2}(a_0a_{n-2} + a_1a_{n-1} + a_2a_n)$$

$$f(x) + f\left(\frac{1}{x}\right) = a_nx^n + a_{n-1}x^{n-1} + x^{n-2}a_{n-2} + \ldots\ldots + a_1x + a_0$$

$$+\frac{a_n}{x^n} + \frac{a_{n-1}}{x^{n-1}} + \frac{a_{n-2}}{x^{n-2}} + \ldots\ldots + \frac{a_1}{x} + a_0$$

$a_n = 1$ $a_0 = 1$; $\{a_1 = a_2 = a_3 = \ldots\ldots = a_{n-1} = 0\}$

$f(x) = x^n + 1$

$f(10) = 10^n + 1 = 10^3 + 1$

$n = 3$

$f(x) = x^3 + 1$

$f(20) = (20)^3 + 1 = 8001$

2. **Ans (A)**

$h(x) = \log(f(x).\,g(x)) = \log e^{\{y\}+\{y\}} = \{y\} + [y]$

$= e^{|x|}\,\text{sng}\,x$

$$\therefore h(x) = e^{|x|}\sin x = \begin{cases} e^x, & x < 0 \\ 0, & x = 0 \\ -e^x, & x < 0 \end{cases}$$

$$\Rightarrow h(-x) = \begin{cases} e^{-x}, & x < 0 \\ 0, & x = 0 \\ -e^x, & x > 0 \end{cases} \Rightarrow h(x) + h(-x) = 0 \text{ for all } x.$$

3. **Ans (C)**

Let co-ordinates of foot of perpendicular are (a, b, c)

Since $\dfrac{a-11}{10} = \dfrac{b+2}{-4} = \dfrac{c+8}{-11} = k$

Then $a = 10k + 11$, $b = -4k - 2$, $c = -11k - 8$

Here all alternative give $a = 1$, then $k = -1$. So foot is (1,2,3) and perpendicular distance between (1,2,3) and (2, –1, 5) is $\sqrt{14}$.

4. **Ans (B)**

Equation of chord of contact

$xh - yk - 9 = 0$ ....(1)

$x - 9 = 0$ ....(2)

$h = 1$ and $k = 0$

pair of tangent

$SS' = T^2$

$(x^2 - y^2 - 9)(-8) = (x - 9)^2\ 9x^2 - 8y^2 - 18x + 9 = 0$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,D)**

(A) $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + ...+{}^nC_nx^n$

Put $x = \omega$ where $\omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$

$({}^nC_0 + {}^nC_3 + {}^nC_6 + ...) + ...+({}^nC_1 + {}^nC_4 + {}^nC_7 + ...)\omega + ({}^nC_2 + {}^nC_5 + {}^nC_8 + ...)\omega^2 = (1 + \omega)^n$

$({}^nC_0 + {}^nC_3 + {}^nC_6 + ...) + ({}^nC_1 + {}^nC_4 + {}^nC_7 + ...)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) + ({}^nC_2 + {}^nC_5 + {}^nC_8 + ...)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$

$({}^nC_0 + {}^nC_3 + {}^nC_6 + ...) - \frac{1}{2}\ ({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + ...) + \frac{i\sqrt{3}}{2}\ ({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + ...) = (-\omega^2)^n$

(B) $a^2b^2 = 4 \Rightarrow 5\log_2(a) + 2\log_2(b) = 2$

$y = 10\log_2 a \cdot 4\log_2 b = 40\log_2 a \cdot \log_2 b$

$A.M. \geq G.M. \Rightarrow \frac{5\log_2(a) + 2\log_2(b)}{2} \geq \sqrt{10\log_2 a.\ \log_2 b}$

$\log_2 a.\ \log_2 b \leq \frac{1}{10} \Rightarrow y \leq 4, y = 4$ when $5\log_2 a = 2\log_2 b \Rightarrow a^5 = b^2 = 2$

(C) For constant term put $x = 0$ and get constant term $= 4$

(D) Coefficient of $x^{24} = 1.25 + 2.24 + 3.23 + ... + 25.1 = 2925$

6. **Ans (A,B,C)**

Let R be the common ratio of the G.P. and D be the common difference of A.P.

$a_5 = a_5$, $a_9 = Ra_5$, $a_{16} = a_5R^2$

$a_9 - a_5 = 4D \Rightarrow (R - 1)a_5 = 4D$

$a_{16} - a_9 = 7D \Rightarrow R(R - 1)a_5 = 7D$

From equation (1) / (2), we get $\frac{1}{R} = \frac{4}{7} \Rightarrow R = \frac{7}{4}$

From equation (2) − (1), we get $(R-1)^2\ a_5 = 3D$

$\Rightarrow \frac{9a_5}{16} = 3D$

$\frac{3}{16}(a_1 + 4D) = D \Rightarrow \frac{3}{16}a_1 = \left(1 - \frac{3}{4}\right)D \Rightarrow a_1 = \frac{4D}{3}$

7. **Ans (A,D)**

$\frac{2x}{(x-1)(x-4)} = \frac{C}{x-1} + \frac{D}{x-4}$

$2x = C(x - 4) + D(x - 1)$

$\therefore C = -2/3, D = 8/3$

$\therefore \int \frac{e^{x-1}}{(x-1)(x-4)}2x\ dx = \int e^{x-1}\left(\frac{-2/3}{x-1} + \frac{8/3}{x-4}\right)dx$

$= -\frac{2}{3}F(x-1) + \frac{8}{3}e^3F(x-4) + C$

$\therefore A = -2/3, B = 8/3\ e^3$

8. **Ans (B,C,D)**

Let $\hat{i},\hat{j},\hat{k}$ be the unit vectors along OA, OB, OC.

CN, BM are diagonals skew to OA

Let 1 be the S.D between OA and CN and let

$\overrightarrow{OA} = x\hat{i}, \overrightarrow{OB} = y\hat{j}, \overrightarrow{OC} = z\hat{k} \Rightarrow \overrightarrow{ON} = x\hat{i} + y\hat{j}$

Equation of OA and CN arfe $\vec{r} = x\hat{i}$ and

$\hat{r} = z\hat{k} + p\left(x\hat{i} + y\hat{j} - z\hat{k}\right)$ respectively.

$\therefore$ The shortest distance between those two lines.

$$l = \frac{yz}{\sqrt{y^2+z^2}} \Rightarrow \frac{1}{l^2} = \frac{1}{y^2} + \frac{1}{z^2}$$

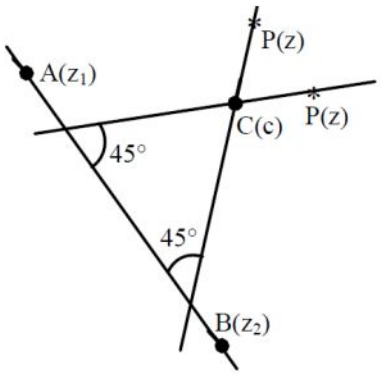
Similarly $\frac{1}{m^2} = \frac{1}{z^2} + \frac{1}{x^2}$, $\frac{1}{n^2} = \frac{1}{x^2} + \frac{1}{y^2}$,

$$\Rightarrow \frac{1}{l^2} + \frac{1}{m^2} + \frac{1}{n^2} = 2\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)$$

9. **Ans (A,B,C)**

Let P(z) be any point on the required line.

Then, $\frac{\vec{CP}}{\left|\vec{CP}\right|}$ i. e. $\frac{z-c}{|z-c|}$ is a unit vector parallel to it



Let $A(z_1)$ and $B(z_2)$ be two points on

$\bar{a}z + a\bar{z} + b = 0$ then $\frac{z_2 - z_1}{|z_2 - z_1|}$ is a unit vector parallel to the line

$a\bar{z} + \bar{a}z + b = 0$

$$\frac{z-c}{|z-c|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{\pm\left(\frac{\pi}{4}\right)}$$

$$\frac{(z-c)^2}{(z-c)(\bar{z}-\bar{c})} = \frac{(z_2 - z_1)}{(z_2 - z_1)(\bar{z}_2 - \bar{z}_1)} e^{\pm i\frac{\pi}{2}}$$

$$\frac{z-c}{z-c} = \pm i\frac{(z_2 - z_1)}{(z_2 - \bar{z}_1)}$$

$\because$ $A(z_1)$ and $B(z_2)$ are on the line

$a\bar{z} + \bar{a}z + b = 0$ therefore $a\bar{z}_1 + \bar{a}z_1 + b = 0$

$a\bar{z}_2 + \bar{a}z + b = 0$

$$\Rightarrow -\frac{a}{\bar{a}} = \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$$

From equation (1) and (2) we get

$$\frac{z-c}{a} \pm \frac{i(\bar{z}-\bar{c})}{\bar{a}} = 0$$

10. **Ans (A,B,C,D)**

$f(x) = x^{1/3}(x-1)$

$\Rightarrow \frac{df(x)}{dx} = \frac{4}{3}x^{1/3} - \frac{1}{3}.\frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}}[4x-1]$

$\Rightarrow$ f(x) changes sign from –ve to +ve, x = 1 / 4, which is point of minima

Also f(x) does not exist at x = 0 as f(x) has vertical tangent at x = 0.

$f''(x) = \frac{4}{9}.\frac{1}{x^{2/3}} = \frac{1}{3}.\frac{2}{3}.\frac{1}{x^{5/3}}$

$= \frac{2}{9x^{2/3}}\left[2+\frac{1}{x}\right] = \frac{2}{9x^{2/3}}\left[\frac{2x+1}{x}\right]$

$\because$ $f''(x) = 0$ at $x = -\frac{1}{2}$ which is the point of inflection at x = 0, f''(x) does not exists but f''(x) changes sign, hence x = 0 is also the point of inflection.

From the information the graph of y = f(x) is as shown.

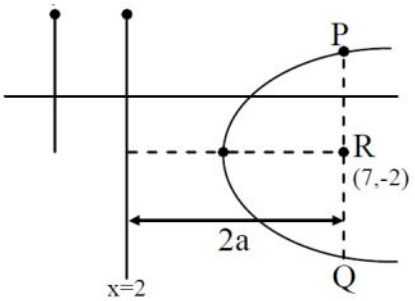
PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (25.00)**

$(x-2) = \sqrt{(x-7)^2+(y+2)^2}$

It is parabola



$(PR)(RQ) = 4a^2$

$= 4 \times \left(\frac{5}{2}\right)^2$

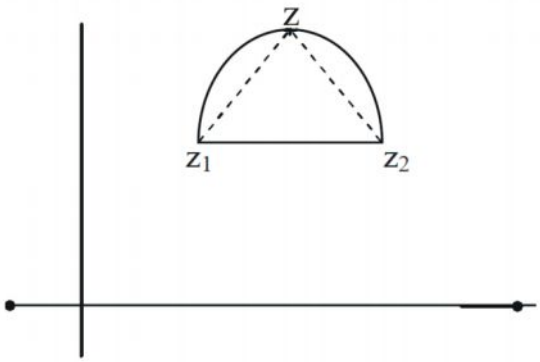
$= 25$

2. **Ans (1.57)**

$Re z - 2 = |z - 7 + 2i|$

Let $P(z_1)$, $Q(z_2)$

$\arg\left(\frac{z_1-(2+\propto i)}{z_2-(2+\propto i)}\right) = \frac{\pi}{2}(\propto \varepsilon R)$



3. **Ans (0.86)**

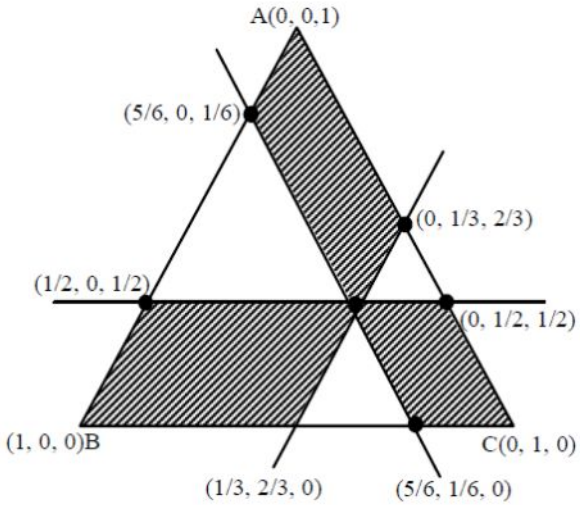
T is an equilateral triangle with the vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1)

$\Rightarrow$ area of the region T is $\frac{\sqrt{3}}{2}$.

4. **Ans (0.34)**

Take a point P $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ on the plane

$x + y + z = 1$

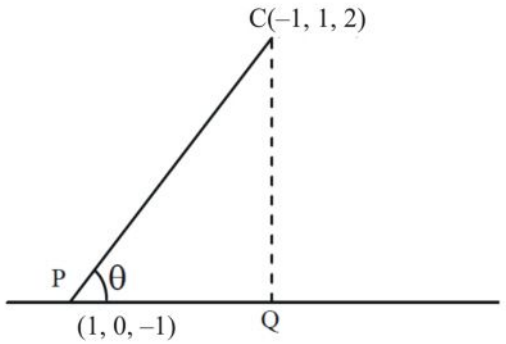


The region S is shown as shaded region.

Area of S = $\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4}\left(a^2+b^2+c^2\right)$, where a, b, c are sides of the small equilateral triangles = $\frac{7\sqrt{3}}{36}$.

6. **Ans (0.45)**

$$\frac{x-2}{z} = \frac{y}{-1} = \frac{z+1}{2}$$



$CQ = CP \sin\theta$

$$= \frac{\left|\vec{CP} \times \vec{L}\right|}{\left|\vec{L}\right|}$$

$$= \frac{\left\| \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -3 \\ 2 & -1 & 2 \end{matrix} \right\|}{\sqrt{4+1+4}} = \frac{\sqrt{125}}{3}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (2)**

E = even of any one cutting a space in one cut

$n(E) = {}^{13}C_1$

$n(S) = {}^{52}C_1$

$P(E) = 1/4 = p$, $P\left(\bar{E}\right) = q$

Probability of a winning = $p + qqqp + qqqqqqp + \ldots\ldots\ldots\infty$

$$= \frac{P}{1-q^3} = \frac{64}{175} = 128$$

$$\left[\frac{\lambda}{64}\right] = 2$$

8. **Ans (8)**

$$f(x) \begin{cases} ax^2 + bx & \text{for } -1 < 1x < 1 \\ \frac{a-b-1}{2} & x = -1 \\ \frac{a+b+1}{2} & x = 1 \\ \frac{1}{x} & \text{for } x > 1 \text{ or } x < -1 \end{cases}$$

for continuity at x = 1 we have $a + b = \frac{a+b+1}{2}$

hence, $a + b = 1$ .....(1)

for continuity at $x = -1$

$a - b = -1$ $a - b = -1$ .....(2)

hence a = 0 and b = 1

9. **Ans (7)**

$P(A \cap B) = P(A)\,P(B) \Rightarrow 1/6 = P(A)P(B)$

$P(A \cup B) = P(A) + P(B) \cdot P(A \cap B)$

$$\Rightarrow 2/3 = P(A) + \frac{1}{6P(A)} - \frac{1}{6}$$

$\Rightarrow 6(P(A))^2 - 5P(A) + 1 = 0 \Rightarrow P(A) = 1/2$, $P(B) = 1/3$

So, $8P(A) + 9P(B) = 4 + 3 = 7$